



MACHINE AND DEEP LEARNING FOR CONFLICT EVENT FORECASTING

MULTI-HORIZON SPATIOTEMPORAL PREDICTION
OF DRONE STRIKES ON UKRAINIAN REGIONS

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Abstract

The Russo-Ukrainian war has made kamikaze drones a defining instrument of modern warfare, motivating the need for forecasting tools that can anticipate where strikes will occur. This thesis evaluates the extent to which machine learning and deep learning algorithms can produce multi-horizon spatiotemporal forecasts of Russian-state-initiated drone-strike events across Ukrainian regions. Daily strike counts were compiled for twenty regions over 846 days (28-09-2022 to 21-01-2025) from the ACLED database and combined with 115 covariates spanning territorial control, weather and geomagnetic activity, cyber and diplomatic signals, international aid announcements, missile-launch counts, and macroeconomic indicators. Gradient-boosted trees (CatBoost, LightGBM, XGBoost) and recurrent networks (LSTM, GRU) were trained under Poisson and Tweedie objectives, and evaluated alongside a time-series foundation model (Chronos-2) and a separate two-stage hurdle pairing a Self-Paced Ensemble classifier with a CatBoost-Tweedie count head. Models were compared via expanding-window backtests against naive, seasonal-naive, and ARIMA baselines over a 1-to-7-day multi-horizon forecast. The top configurations cluster within a narrow band (Skill Score 0.15-0.17), with a univariate ARIMA baseline within 0.02 root mean squared error of the leaders — indicating that recent autoregressive history captures most of the predictable variance, especially in high-activity regions. The hurdle architecture underperforms a single-stage Tweedie objective, RMSE degrades significantly beyond a four-day horizon while mean absolute error remains stable, and meteorological and autoregressive features dominate the importance rankings.

1 SOURCE / CODE / ETHICS / TECHNOLOGY STATEMENT

Data Source, Ethics, Code, and Technology Statement

Data Source: All data used in this thesis has been collected from the sources described in Section 4.1 through online requests. The datasets include: the Armed Conflict Location and Event Data project (ACLED), the VIINA territorial control dataset, Open-Meteo historical weather data, NOAA daily geomagnetic data, the Cyber Events database, the Ukraine Support Tracker (Kiel Institute), the GDELT events database, the Massive Missile Attacks on Ukraine dataset, the Nuclear Escalation Risk Signaling dataset, exchange rate data via the Yfinance Python package and Holiday data retrieved via the Python holidays package. All data collected originates from open-source and open-access sources that permit usage for scientific and research purposes. This thesis did not involve collecting data from human participants or animals. The original owners of all datasets retain ownership of their respective data during and after the completion of this thesis. Figures: All figures included in this thesis were created by the author. Code: All code produced for the experiments in this thesis is publicly accessible via the GitHub repository at <https://github.com/J4nvg/thesis>. Technology: During the preparation of this work, the author used a thesaurus (<https://www.thesaurus.com/>) and large language models (Anthropic. (2026). *Claude Opus 4.7* [Large language model]. <https://claude.ai>; Google. (2026). *Gemini 3.1 pro* [Large language model]. <https://gemini.google.com>) to iteratively improve the author's original content, for paraphrasing, spell checking and grammar, code review, and academic correctness of the thesis.

2 INTRODUCTION

The Russo-Ukrainian war, which escalated with the Russian invasion in early 2022, has drastically altered the role and scope of modern warfare (Veilleux-Lepage & Rassler, 2025). In particular, unmanned aerial vehicles—notably kamikaze drones—have become smaller, more accessible, providing a highly lethal and affordable alternative to classical long-range artillery (Molloy, 2024). These drones pose a severe human threat and are frequently deployed to degrade critical infrastructure, including energy, health, and educational facilities. The international dimensions of the conflict further complicate this landscape, as Russia receives material support from states like Iran, China, and the Democratic People's Republic of Korea, while Ukraine relies on backing from Western allies (Trebesch et al., 2023; Veilleux-Lepage & Rassler, 2025).

Simultaneously, the widespread use of smartphones has generated an unprecedented volume of open-source intelligence, transforming how conflicts are documented in real-time. To illustrate, non-Governmental Organizations collected 2.8 million digital records of potential war crimes

in Ukraine between February and May 2022 alone, compared to 5 million records gathered over eleven years in the Syrian conflict (Ford, 2024). The massive influx of real-time data has catalyzed the use of machine learning for forecasting conflict-related events (Murphy, Sharpe, & Huang, 2024).

Predictive modelling is becoming increasingly integral to military decision-making (Baboş, Iacob, & Ene, 2022), and applying it to drone warfare presents a critical opportunity: forecasting affected regions in advance can mitigate damage, minimize casualties, and ensure timely resource distribution. The value of the forecast, however, depends on how far ahead it remains accurate. A single lead time is rarely sufficient operationally, since shorter and longer warnings serve different decisions. The forecasting target is therefore multi-horizon: strike counts are predicted jointly at each lead time from one up to seven days ahead, rather than at one fixed forecast horizon.

Equally open is which algorithm class best fits this setting. Machine-learning and deep-learning forecasters contain opposing inductive biases: gradient-boosted trees excel on tabular, feature-rich data but cannot extrapolate beyond their training range, whereas recurrent neural networks and pretrained time-series foundation models model sequential structure natively, yet demand more data for training (Ansari et al., 2025; Hall & Rasheed, 2025; Lim & Zohren, 2021).

Which performs better on relatively short, sparse, zero-inflated conflict series at a daily, sub-national resolution remains comparatively underexplored: although both families have been featured in conflict-forecasting challenges, they are seldom co-evaluated head-to-head within a single controlled pipeline (Hegre et al., 2025; Murphy et al., 2024). Resolving this empirically – and identifying which signals drive the forecasts—this thesis compares deep- and machine-learning algorithms for the spatiotemporal forecasting of Russian-state-initiated drone-strike events in Ukrainian regions.

Research Question:

To what extent can deep-learning and machine-learning algorithms be utilized for multi-horizon spatiotemporal forecasting of Russian-state initiated drone-strike events in Ukrainian regions?

To systematically address this research question, the following sub-questions have been formulated:

- Sub-Q1 *How do machine learning methods compare against deep-learning based methods in this context?*
- Sub-Q2 *How does the 1-to-7-day multi-horizon setting impact the forecast performance?*

Sub-Q3 *For top performing models, what are the most important predictive features?*

Across twenty Ukrainian oblasts and a 1-to-7-day multi-horizon setting: gradient-boosted trees, recurrent networks, and a fine-tuned time-series foundation model cluster within a narrow performance band. A univariate ARIMA baseline lies within 0.02 root mean square error (RMSE) of the leaders, indicating that recent autoregressive history captures most of the predictable variance. Typical-day forecast error remains stable across the horizon while RMSE degrades significantly beyond four days. Moreover, a hurdle architecture underperforms a single-stage Tweedie objective, and meteorological and autoregressive features dominate the importance rankings.

3 RELATED WORK

Over the past two decades, machine learning for conflict-event prediction has shifted from an exploratory pursuit to an actionable forecasting objective, combining event-based signals such as armed clashes with slower structural drivers like economic and climate indicators (Murphy et al., 2024), and is increasingly central to military decision-making under uncertainty (Baboş et al., 2022). The Violence & Impacts Early-Warning System (VIEWS), an academic research consortium, exemplifies this trend: its 2023/24 challenge frames the prediction of state-based conflict fatalities over a 12-month window as a time-series problem, drawing contributions that span gradient-boosted tree ensembles, transformer architectures, and hurdle compositions, all under a range of count-distributional assumptions (Hegre et al., 2025).

Conflict-forecasting models track ongoing violence reasonably well, but predict the rare onsets and escalations less reliably (Bazzi et al., 2022; Hegre, Nygård, & Landsverk, 2021). This limitation likely finds its cause in how conflict data behave, and it is what sets the task apart from smoother forecasting problems. Analysing African ACLED events, Lee, Daniels, Myers, Krakauer, and Flack (2020) show that armed conflict unfolds in avalanche-like cascades whose fatalities, durations, and spatial extents follow heavy-tailed, power-law distributions, so that a few samples are vastly larger than the rest. The same heavy right-tail recurs in war-casualty totals (Cirillo & Taleb, 2016) and in the state-based fatality series that early-warning systems target (Hegre et al., 2025). The sample maximum is therefore an unreliable ceiling, as future escalation can exceed anything observed during training. This highlights the importance of a model's capacity to extrapolate beyond its training support.

Across the literature the conflict-prediction target is usually aggregate: fatality counts or conflict risk, and even rigorous subnational assessments find *where* violence occurs more tractable than *when* (Bazzi et al., 2022). A specific violent event type, or particular weapon platform, is rarely the prediction target. This likely reflects both the rarity of such events and the relative scarcity of accessible standardised event-level data compared with the aggregate fatality and risk series. Found literature about the present conflict shows that – despite the volume of data – much of the published machine learning work clusters around sentiment analysis of social-media discourse (Aslan, 2023; Vyas, Vyas, & Dhiman, 2023; Wadhvani, Varshney, Gupta, & Kumar, 2023) or macroeconomic and financial impact modelling (Dobrovolska, Kolotilina, Ostapenko, et al., 2024; Mati et al., 2023; Sliusarenko, Baranov, & Voytsehovskiy, 2024). Explicit violent events, such as drone strikes, remain largely unmodelled as a time-series forecasting problem.

Gradient boosting frameworks, most notably LightGBM, CatBoost, and XGBoost, have shown reliable in time-series forecasting, prized for their computational efficiency, robustness to redundant features, and interpretability (Aydın, Erdem, & Cicek, 2023; Teusch et al., 2023). LightGBM in particular emerges as a consistent top performer on tabular time-series benchmarks (Hall & Rasheed, 2025), and the leading solutions in the M5 forecasting competition were almost exclusively LightGBM-based — with the winning entry employing a Tweedie loss to accommodate zero inflation (Makridakis, Spiliotis, & Assimakopoulos, 2022). A fundamental limitation, however, is the inability of tree-based learners to extrapolate beyond the empirical support of their training data. This poses a structural constraint of particular concern in evolving conflict environments, where escalation dynamics may push observed counts beyond historically attested ranges.

To capture temporal dependencies that tree-based models represent only indirectly through lagged features, deep learning architectures are widely adopted in time-series forecasting. Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), and Convolutional Neural Networks (CNNs) constitute the dominant families (Lim & Zohren, 2021; Mendis, Wickramasinghe, & Marasinghe, 2024), with RNN variants demonstrating notable effectiveness even on relatively small datasets (Hall & Rasheed, 2025). LSTMs are particularly well-suited to the present task: they natively model sequential structure and, unlike tree-based learners, are in principle capable of extrapolating beyond the training distribution, making them a natural complement to the gradient boosted models.

A more recent development is the emergence of foundation models for time-series forecasting. Whereas earlier general-purpose forecasters such

as Prophet (Taylor & Letham, 2018) and TimesFM (Das, Kong, Sen, & Zhou, 2024) were designed primarily for univariate input, the recently released Chronos-2 natively accommodates multivariate series and is engineered for zero-shot prediction with optional fine-tuning (Ansari et al., 2025). Such models are attractive in the data-scarce regimes characteristic of emerging conflicts, where transferable representations may compensate for limited task-specific history. Murphy et al. (2024) explicitly highlight foundation models as a promising direction for conflict forecasting, motivating their inclusion in the comparative analysis.

Because drone strike events are rare in most regions, the target distribution is dominated by zeros that standard count models cannot accommodate without under-predicting the rare high-count events. The statistical literature has long contended with such excess zeros, yielding two principal modelling traditions. Zero-Inflated Poisson (ZIP) regression (Lambert, 1992) treats observed zeros as a mixture of structural and sampling zeros, while hurdle models (Cragg, 1971; Mullahy, 1986) decompose the data-generating process into two stages: a binary component governing event occurrence, and a zero-truncated count component governing magnitude conditional on occurrence. Hurdle models have since been adapted to machine learning settings (Ben Khedher & Yun, 2024; Rožanec et al., 2025; Xu, Ye, Gao, & Chu, 2024), while zero-inflated regression has also been addressed within a single model through specialized loss functions for gradient boosted trees, including the zero-inflated Poisson (So, 2024) and Tweedie (Zhou, Qian, & Yang, 2022) objectives.

The hurdle formulation is particularly attractive here for two reasons. First, it permits different inductive biases for the two sub-problems, recognizing that the determinants of *whether* a strike occurs in a region may differ from those governing *how many* occur. Second, it makes the severe class imbalance inherent to rare-event forecasting an explicit modelling concern. The binary stage thereby reduces to a highly imbalanced classification task, for which a substantial methodological literature spans resampling, ensembling, and synthetic data generation (W. Chen, Yang, Yu, Shi, & Chen, 2024). The recent Class-Imbalanced Learning Benchmark on Tabular Data (CLIMB) systematically compared these approaches and identified the Self-Paced Ensemble (SPE) (Liu et al., 2020) as the strongest overall performer (Liu et al., 2025), motivating its adoption as the classification component of the hurdle architecture.

This thesis addresses two gaps. Empirically, machine learning on the Russo-Ukrainian war has concentrated on peripheral digital signals—social-media sentiment, macroeconomic volatility—leaving the spatiotemporal

distribution of the kinetic events themselves unmodelled. Methodologically, model families like these have already co-occurred on conflict data—the VIEWS 2023/24 challenge alone spans tree ensembles, transformers, and two-stage hurdle models on a shared fatality target (Hegre et al., 2025)—but as separate submissions at monthly, aggregate resolution, not a controlled comparison. It remains uncharacterised how these families—single-stage or hurdle—compare under a single shared pipeline at the daily, regional-level, single event type studied here.

4 METHODOLOGY

4.1 Data collection

The Armed Conflict Location and Event Data project (ACLED), introduced in 2010 (Raleigh, Linke, Hegre, & Karlsen, 2010), is a research standard for tracking conflict-related events globally (Murphy et al., 2024). This researcher-led project draws on a dynamic network of multilingual local partner sources, prioritising external validity (Raleigh, Kishi, & Linke, 2023). Having provided structured, event-based data since the onset of the Russo-Ukrainian war¹, it was selected as the primary source for historical drone-strike events in Ukrainian regions. Each event carries an extensive feature set, including `event_date`, `disorder_type`, `event_type`, `sub_event_type`, `actor1`, `actor2`, `country`, `admin1`, and `note`, defined in Table 1 after the ACLED codebook (Armed Conflict Location & Event Data Project (ACLED), 2026).

Table 1: Description of primary ACLED features for final dataset.

Feature	Description
<code>event_date</code>	The specific date on which the strike event was recorded.
<code>disorder_type</code>	The broad category of disorder (e.g., Political violence, Strategic developments).
<code>event_type</code>	The specific nature of the event, typically "Explosions/Remote violence" for strikes.
<code>sub_event_type</code>	The finest level of categorization, such as "Air/drone strike."
<code>actor1</code> / <code>actor2</code>	The primary and secondary entities involved in the event.
<code>admin1</code>	The highest-level sub-national region (Oblast) where the event occurred.
<code>note</code>	A short description providing qualitative context (e.g., specific munitions used or targets hit).

¹ <https://acleddata.com/monitor/ukraine-conflict-monitor>

Following [Mostert et al. \(2025\)](#), the ACLED data were extended with contextual covariates. That study's codebook and open-source conflict reporting guided the identification of potentially informative variables, yielding a feature set that spans region-specific indicators and measures of international interaction. This feature set aims to capture the conflict's geopolitical and behavioral patterns, with the goal of maximizing predictive accuracy and identifying key drivers of strike events. Reflecting the multiple international actors noted in Section 2, the feature set captures interactions among the primary state actors (Russia and Ukraine) and secondary actors, both with the primaries and among themselves. Secondary state actors include NATO, the European Union, the United States, Iran, Belarus, Central Asia, and the DPRK.

A separate ACLED dataset tracking Ukrainian civilian infrastructure damage was consulted. This dataset retains the core features outlined in Table 1, but adds a variable categorizing the type of infrastructure affected.

The VIINA dataset ([Zhukov & Ayers, 2023](#)) contains a subset tracking territorial control in Ukrainian regions over time. This component was used to calculate the following daily indicators:

- The distance from the centroid of each region to the centroid of the nearest polygon designated as Russian-claimed territory (*km*).
- The distance from the centroid of each region to the centroid of the nearest polygon designated as contested territory (*km*).
- The total surface area of Russian-claimed territory (*km*²).
- The total surface area of contested territory (*km*²).

Open-Meteo ([Zippenfenig, 2023](#)) provided historically accurate regional daily weather metrics, motivated by the documented influence of weather on UAS and counter-UAS performance ([Sobolewski, 2025](#)). As most UAS depend on radio or GPS, the daily maximum planetary k-index — a standard measure of geomagnetic activity known to disrupt radio communications ([Balaji & Ranganathan, 2024](#)) — was retrieved from the NOAA (National Centers for Environmental Information & Space Weather Prediction Center, n.d.).

The cyber-events database ([Harry & Gallagher, 2018](#)) provides a systematic log of dates, targets, and initiators of cyber-attacks. From this dataset, daily counts of bi-directional primary-primary and primary-secondary cyber interactions were extracted.

The Ukraine Support Tracker from the Kiel Institute ([Trebesch et al., 2023](#)) records announcement dates for different categories of international support provided by western countries. Daily announcement counts were

extracted for the following aid types: Financial Aid, Humanitarian Aid, and Military Aid.

The GDELT events database (Leetaru & Schrodt, 2013) was utilized for the daily extraction of signals from news outlets related to verbal conflict, diplomatic agreement, reduction of relations, sanctions, escalation signals, and logistical cooperation, covering all primary and secondary state actors involved in the conflict. Signal extraction was based on a predefined set of CAMEO codes.

Massive Missile Attacks on Ukraine (Ivaniuk, 2026) is a dataset compiled by developer Petro Ivaniuk. The data were verified by the Center for Strategic and International Studies (CSIS) against official figures published by the Ukrainian Air Force (Atalan, n.d.). From this dataset, daily counts of confirmed launched and intercepted drones were extracted.

Nuclear Escalation Risk Signaling The dataset introduced by (Horovitz et al., 2025) provides specific nuclear escalation and de-escalation signals attributed to Western governments, Russia, and Belarus, including statements of affirmatory deterrence and formal policy adjustments.

The **Yfinance** Python package (Aroussi, n.d.) was utilized to retrieve historical daily exchange rates for the Russian and Ukrainian currencies against the US Dollar to capture regional macroeconomic fluctuations. The **Holidays** Python package (Boogaard, McKinney, & contributors, 2026) was used to retrieve accurate indication of Russian and Ukrainian holidays.

See appendix for all variables used: 9.

4.2 Dataset description

Although the majority of the consulted data sources provide observations from the onset of the full-scale invasion on 24-02-2022 (see Figure 1), the *Massive Missile Attacks on Ukraine* dataset (Ivaniuk, 2026) only begins recording confirmed daily counts of launched and intercepted drones on 28-09-2022. Because daily drone-strike events per oblast correlate strongly with the number of drones dispatched, this covariate is indispensable; retaining the seven preceding months would leave a structurally important predictor constantly missing. The observation window is therefore left-truncated at 28-09-2022.

The right-hand boundary is dictated by data-provider lag. ACLED publishes its full event-level archive with an one-year delay; the most recent complete record available at retrieval was 21-01-2025. The final observation window therefore spans **846 days**, from 28-09-2022 to 21-01-2025 (Figure 1).

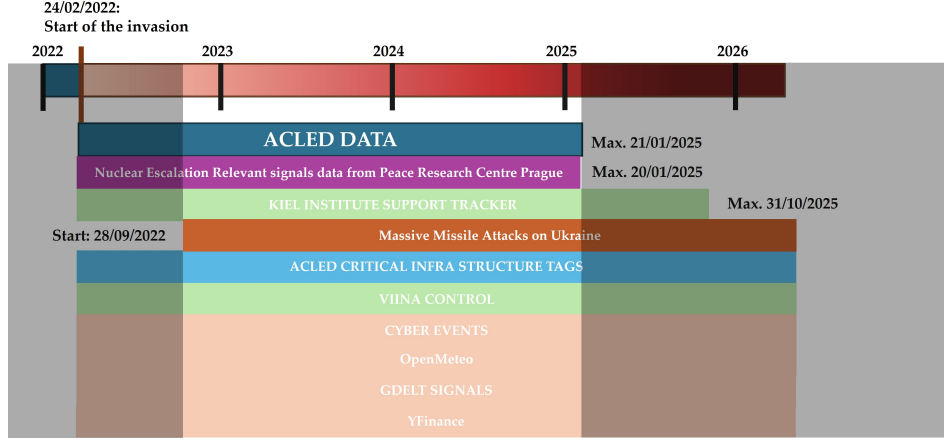


Figure 1: Native temporal coverage of each consulted data source. The shaded interval [28-09-2022, 21-01-2025] indicates the common window retained for modelling.

Drone-strike incidence is highly heterogeneous across Ukrainian territory. Figure 2 reports, for each oblast, the number of unique days on which at least one drone-strike event was recorded during the full 846-day window. The empirical distribution ranges from 0 (*Zakarpattia, Crimea*) to 411 (*Sumy*) and exhibits a clear separation into distinct activity regimes. To make this stratification explicit, each oblast is assigned to one of four *activity tiers* on the basis of its unique-day drone-incidence count n_{days} :

Tier	n_{days}	Description
Tier 0	$n_{\text{days}} < 7$	Regions in which drone-strike incidence is near-zero or exactly zero over the observation window; <i>excluded</i> from experimentation.
Tier 1	$7 \leq n_{\text{days}} < 84$	Regions with a sparse but non-trivial strike history, typically distant from the front line.
Tier 2	$84 \leq n_{\text{days}} < 219$	Regions with regular but moderate exposure, including major urban centres and southern coastal oblasts.
Tier 3	$219 \leq n_{\text{days}} \leq 411$	Front-line and adjacent regions subjected to sustained, high-intensity strike activity.

Table 2: Stratification of activity tiers on the basis of the unique-day drone-incidence count n_{days} .

The tier boundaries are chosen as natural breakpoints in the empirical distribution of unique-day counts rather than by an algorithmic procedure; they enable stratified analysis of forecasting performance across activity

regimes (Section 7) and serve as a static covariate. Figure 3 maps the resulting tier assignment back onto Ukrainian geography.

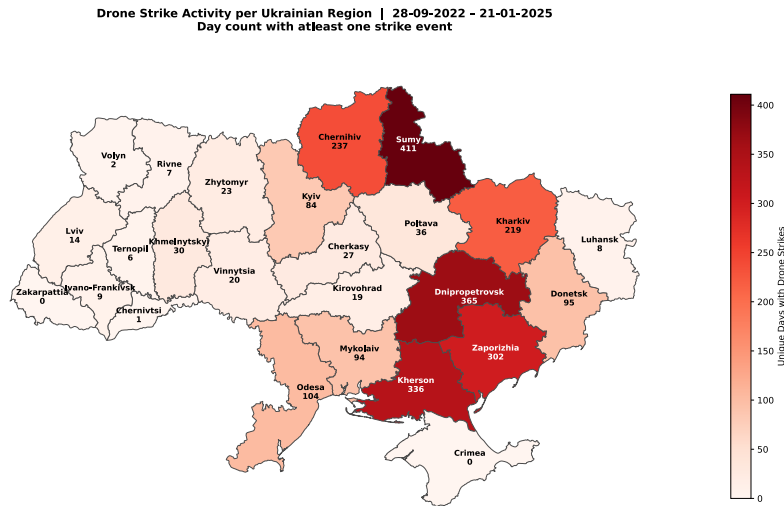


Figure 2: Drone-strike activity per Ukrainian oblast over the period 28-09-2022 to 21-01-2025, measured as the number of unique days with at least one recorded drone-strike event. (Note: Kyiv and the separate administrative region of Kyiv City have been merged into a single geographic entity: Kyiv.)

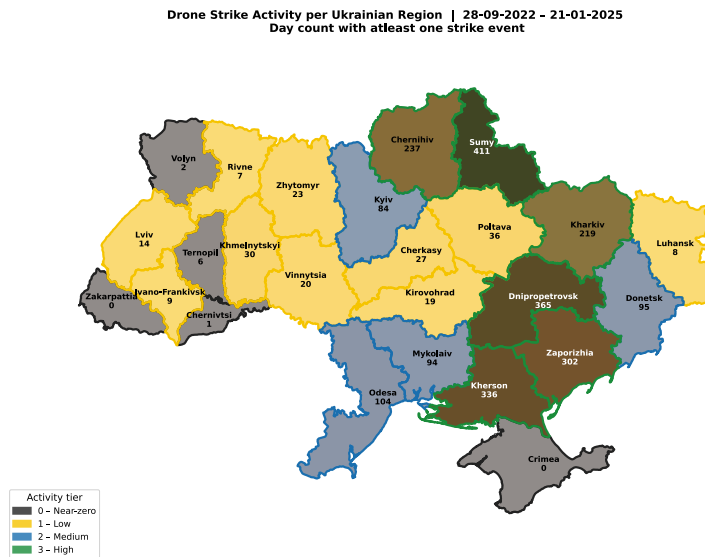


Figure 3: Activity-tier assignment per Ukrainian oblast. Tier-0 regions (grey) are excluded from the modelling dataset.

After excluding the Tier-0 regions, the modelling dataset comprises **20 oblasts** observed over **846 days**, yielding $20 \times 846 = 16,920$ region-day observations. The data are partitioned chronologically into a training, validation, and test set in a 70/10/20 ratio, applied identically to every region so that all splits remain spatially balanced. The resulting split is summarised in Table 3.

Before the generation of cyclic-calendar, signal-propagation, and lag features (Section 4.4), each region-day observation is associated with a covariate vector of dimension 115.

Table 3: Dimensions of the chronological train-validation-test split.

Split	Days / region	Observations
Training	592	11,844
Validation	85	1,692
Test	169	3,384
Total	846	16,920

4.3 Exploratory data analysis

Figure 4 presents the spatio-temporal structure of the target as a weekly intensity heatmap. Two observations stand out. First, weekly intensity grows considerably over time in nearly every Tier-2 and Tier-3 region indicating non-stationarity in the data-generating process. Second, the validation and test windows fall within the interval of highest observed intensity. This makes the forecasting task substantially harder than a uniform chronological split would, though it also yields the most operationally realistic assessment.

Figure 5 characterises the marginal distribution. The top panel confirms strong zero-inflation: frequency drops roughly an order of magnitude per additional event. The bottom panel reports, per activity tier, the zero rate and the index of dispersion $D = \text{Var}(Y)/\mathbb{E}[Y]$, which measures departure from the equi-dispersion property of the Poisson distribution ($D = 1$); values above one indicate over-dispersion (Cameron & Trivedi, 2013). The tiers separate cleanly: Tier 1 is near-Poisson ($D = 1.15$, zero rate 0.98), Tier 2 is moderately over-dispersed ($D = 2.03$), and Tier 3 is strongly over-dispersed ($D = 5.16$). This pattern motivates the Tweedie objective for the count models and the hurdle decomposition of Section 5.4.

Figure 6 reports the autocorrelation and partial autocorrelation (Hyndman et al., 2025) of the national daily total, the latter isolating the direct contribution of lag k after intermediate lags have been partialled out (Joseph & Tackes, 2024). Strong direct dependence is visible at lag 1, a secondary

peak at lag 2, and a recurring spike at lag 7 consistent with weekly seasonality. The ACF fails to decay over thirty lags, a standard indicator of non-stationarity (Hyndman et al., 2025).

Trend strength F_T and seasonal strength F_S (Wang, Smith, & Hyndman, 2006) are computed on an STL decomposition of the national total. Hyndman et al. (2025) prescribe first-differencing to induce stationarity and additionally recommend seasonal differencing whenever $F_S > 0.64$. Figure 7 shows that a single first-difference reduces F_T from 0.73 to 0.09, effectively eliminating the trend, while F_S is essentially preserved (0.38 to 0.36, well below the seasonal-differencing threshold). A single non-seasonal difference is therefore retained as a target transformation in the differenced-target regression branch of Figure 8.

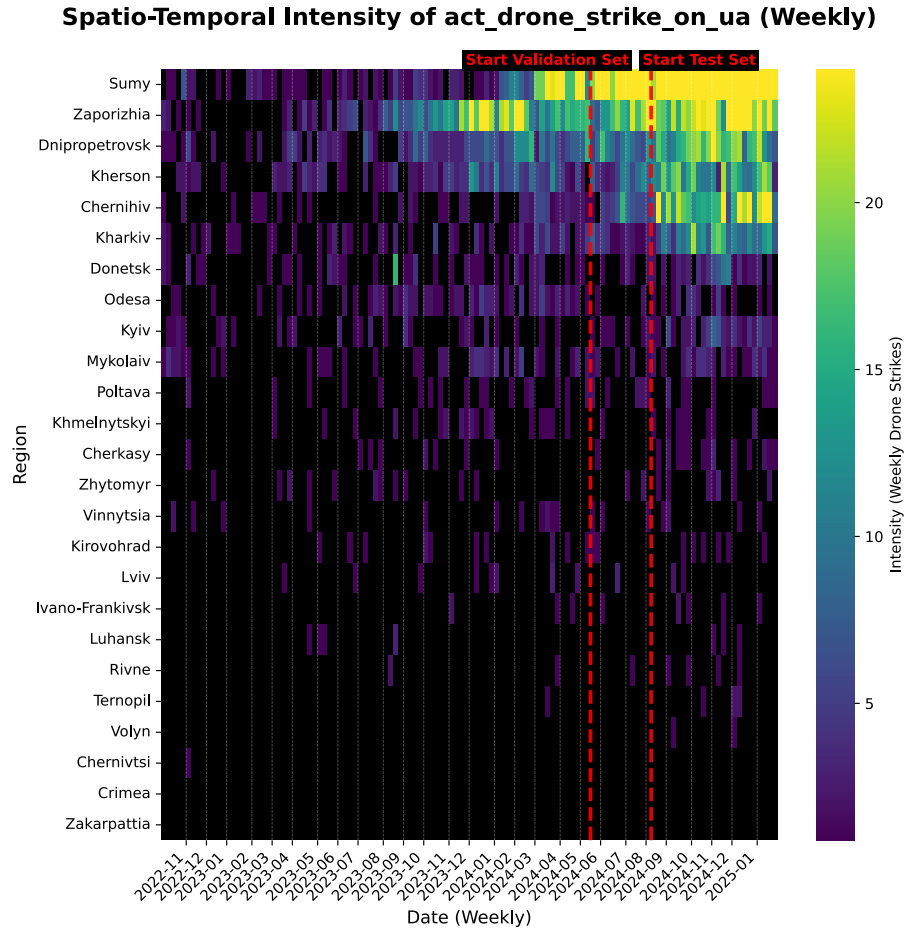


Figure 4: Spatio-temporal heatmap of weekly drone-strike intensity per oblast over the period 28-09-2022 to 21-01-2025. Dashed red lines mark the start of the validation and test partitions, respectively.

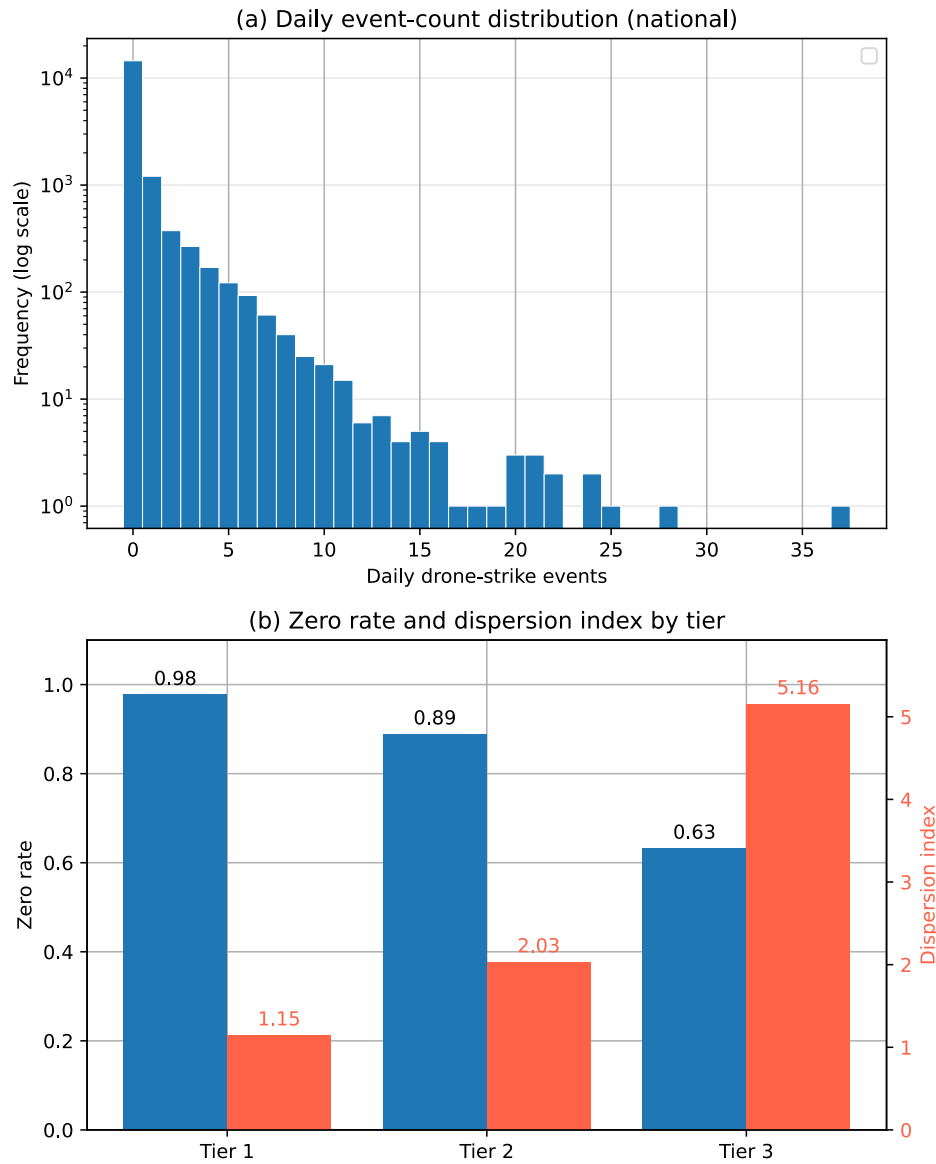


Figure 5: Distribution of the target variable. Left: empirical frequency of daily drone-strike counts per region-day, on a logarithmic frequency axis. Right: zero rate (blue, left axis) and index of dispersion $D = \text{Var}(Y)/\mathbb{E}[Y]$ (red, right axis) by activity tier.

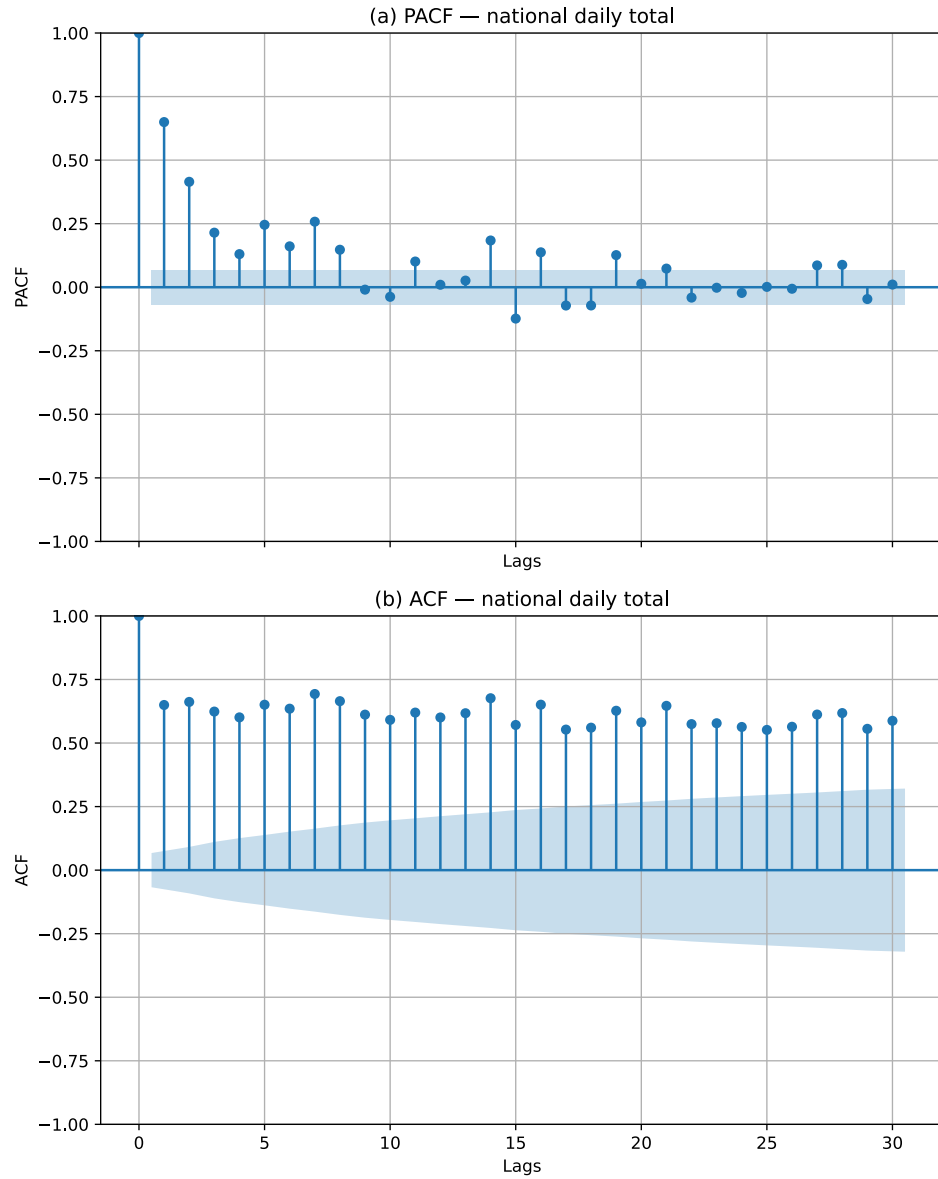


Figure 6: Partial autocorrelation (left) and autocorrelation (right) of the national daily drone-strike total up to lag 30. Shaded bands denote the 95% confidence intervals under the null of no autocorrelation. Highest Partial autocorrelation for the first two, fifth and seventh day.

STL Decomposition: Raw Target vs. Differenced Target

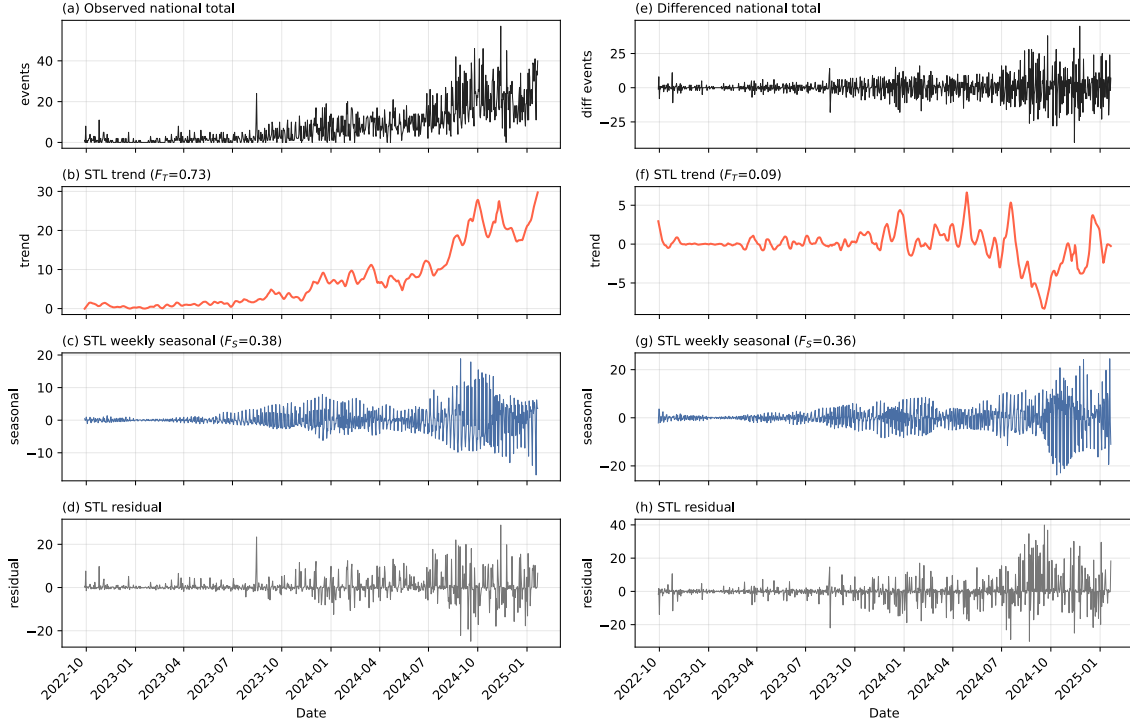


Figure 7: STL decomposition of the national daily total before (left) and after (right) first-differencing. Reported trend strengths F_T and seasonal strengths F_S follow Wang et al. (2006).

4.4 Feature Engineering

Raw conflict and exogenous data rarely carry the structure predictive algorithms require. Feature engineering is therefore applied to expose hidden temporal, spatial, and operational dependencies in these inputs.

The ACLED dataset groups all air-based offensive operations under a single subtype, failing to distinguish manned airstrikes from unmanned attacks, or short-range FPVs from long-range loitering munitions. Because this study targets the latter, the raw label is insufficient.

Therefore, the notes field is parsed in three stages (detailed in Appendix A): preprocessing first neutralizes non-strike mentions (e.g., reconnaissance); regular expressions then tag aviation, drone, and interception language; finally, a combination rule isolates drone strikes by requiring offensive patterns or general mentions without interception context. Airstrike and drone-strike labels remain non-exclusive as a single note can describe

both.

Drone-strike records are then stratified further along up to four dimensions: the initiating party, the country in which the strike occurred, the platform class (FPV versus non-FPV), and— for strikes taking place on Ukrainian territory—the sub-national region in which the event occurred. The first two dimensions are recovered from ACLED’s structured actor and country fields and are treated as reliable. The third is inferred from the notes field: a drone strike is labelled FPV if the term appears in the cleaned note, and non-FPV otherwise. Events occurring in Russia are not further disaggregated by region.

The assumption is made that every FPV-based strike is explicitly identified as such by the ACLED coder. Unfortunately this is an assumption that cannot be verified against ground truth. The resulting non-FPV class should therefore be read as “*not identified as FPV*” rather than as a clean negative. The Cartesian product of the first three dimensions yields six base event categories (e.g. `rus_in_ua_fpv`); the two `_in_ua` families are additionally partitioned by region into oblast-level sub-categories.

The annotated ACLED database introduced in Section 4.1 was first filtered using the same drone-versus-airstrike disambiguation, yielding a subset of events attributable specifically to drone activity. Within this subset, however, a substantive distinction emerges regarding the pathway through which infrastructure damage occurs. To illustrate this distinction, consider the following two excerpts drawn directly from the annotated dataset:

“On 16 May 2025, debris of Russian-kamikaze drone shot down by the Ukrainian military hit a heating infrastructure, residential buildings and cars in Kyiv - Sviatoshynskyi, Kyiv City. There were no casualties.”

Here, damage to heating infrastructure, residential buildings, and vehicles is a secondary consequence of interception; because the drone was neutralised mid-flight, its intended objective cannot be recovered from the data. By contrast, a separate ACLED entry from 10 October 2025 reports Russian drones damaging three residential high-rises, four stores, and vehicles in Brovary with no evidence of interception. In such cases, the affected structures are treated as the intended targets. This dichotomy motivates a per-event classification of damage as either *intentional* or *unintentional*. The classification is done by using the LLM-as-a-judge framework (Gu et al., 2024). See Appendix B for an overview of the system and the used

system-prompt.

To help the algorithms capture recurring temporal patterns—such as weekly operational tempos or seasonal weather shifts—calendar features (month, week, dayofyear, dayofweek, day) are introduced as future-known covariates. To preserve their inherent cyclical continuity without introducing artificial linear boundaries (e.g., the jump from December 31 to January 1), these are encoded into continuous sine and cosine components (Joseph & Tackes, 2024; Tziortas, Theodoropoulos, & Theodoridis, 2026):

$$\text{Feature}_{\sin} = \sin\left(\frac{2\pi \cdot \text{current_value}}{\text{max_value}}\right) \quad (1)$$

$$\text{Feature}_{\cos} = \cos\left(\frac{2\pi \cdot \text{current_value}}{\text{max_value}}\right) \quad (2)$$

As gradient boosted trees lack a mechanism for sequential dependence, discrete raw signals must be explicitly propagated across time to provide temporal context. Following Joseph and Tackes (2024) and Tziortas et al. (2026) Rolling and exponential window transformations are applied to the observation sequence x_t .

Rolling window transformations. Rolling sums and means over a fixed lookback window of size W :

$$\text{rsum}_W(t) = \sum_{i=0}^{W-1} x_{t-i} \quad (3)$$

where $W \in \{7, 14\}$.

$$\text{rmean}_W(t) = \frac{1}{W} \sum_{i=0}^{W-1} x_{t-i} \quad (4)$$

where $W \in \{7, 28\}$.

Exponential window transformations. To propagate discrete signals with decaying weight, two exponential filters are applied. The Exponential Moving Average (EMA) for span S :

$$s(t) = \alpha \cdot x_t + (1 - \alpha) \cdot s(t - 1) \quad (5)$$

where the smoothing factor $\alpha = \frac{2}{S+1}$. The model utilizes an EMA with a span of $S = 14$ ($\alpha \approx 0.133$).

An Exponential Decay (leaky integrator) filter, simulating the lingering effect of discrete events:

$$s(t) = x_t + \alpha \cdot s(t - 1) \quad (6)$$

where the decay factor $\alpha \approx 0.905$ is derived from a defined half-life of 7 days.

To establish explicit autoregressive relationships, strict lagging structures are imposed on the inputs:

- **Target Lags:** The past 7 observations of the target variable ($t - 1$ to $t - 7$) are provided as historical context.
- **Past Covariates:** Selected seasonal and recent lags are provided at specific intervals ($t - 1$, $t - 7$, and $t - 14$).
- **Future Covariates:** Known future features (including the cyclic calendar variables) are provided with a historical lookback of 2 steps and a lookahead spanning the full seven-day forecast horizon ($W_{out} = 7$).

4.5 Feature selection

Following the expansion of the feature space through signal-propagation and lag-feature generation, a pruning step is required to mitigate overfitting on noise. For each target, a preliminary LightGBM model is trained on the full feature set under the same objective as the corresponding downstream models. Since the modelling framework employs independent estimators per forecast step, retained features must be informative across the entire horizon. The native LightGBM gain-based importance is therefore extracted for each horizon day and averaged into a single mean information gain per feature. Features are ranked in descending order of this mean information gain, and the top 100 are retained as the definitive feature space for subsequent experimentation; the procedure is illustrated in Figure 8. Cyclic calendar features and weather variables are exempt from this filter and supplied for the complete horizon (where available), as future-known covariates are considered highly informative.

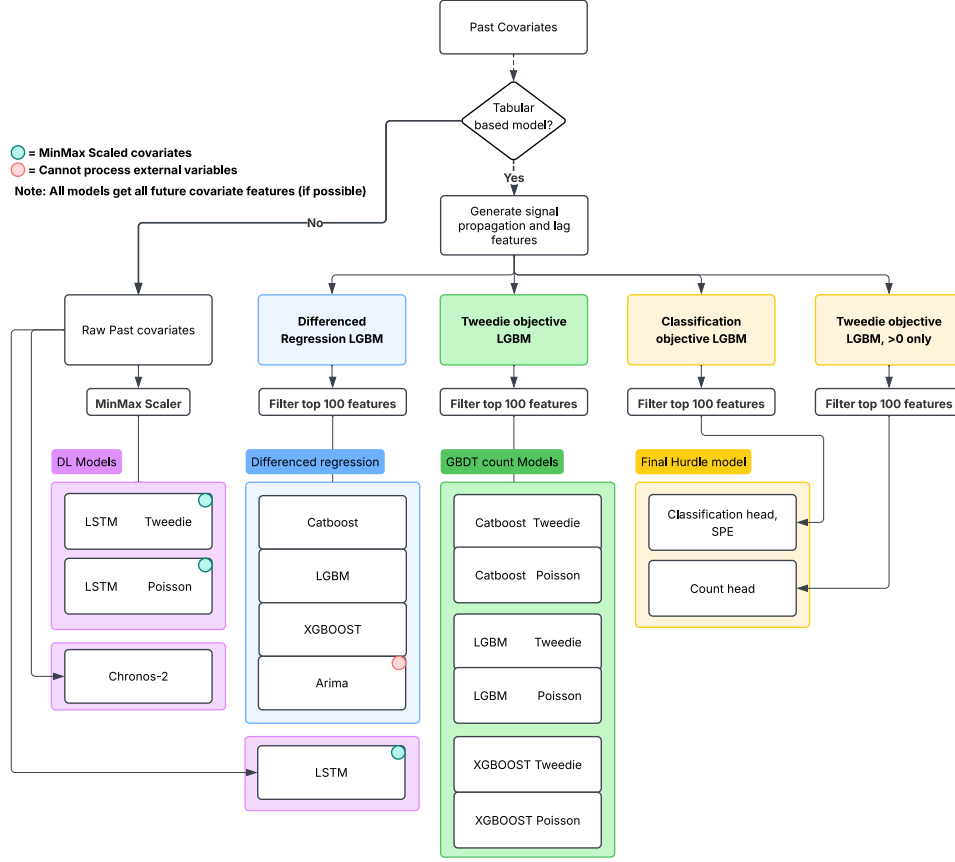


Figure 8: Modelling pipeline for feature-selection procedure. For each tabular branch (differenced/log regression, Tweedie count, classification, Tweedie-on-positives), a preliminary LightGBM model trained under the matched objective ranks features by mean horizon-averaged gain; the top 100 define the feature space for the downstream models in that branch. Deep-learning models receive raw, MinMax-scaled past covariates without lag engineering. Cyan: MinMax-scaled inputs. Red: model cannot ingest exogenous variables.

5 ALGORITHMS

Several baselines are common in the time series domain, among them the naive forecaster, the seasonal naive forecaster, and the Autoregressive Integrated Moving Average (ARIMA) model. The naive baseline relies on the fact that for a random walk process the optimal point forecast is the most recent observation, $\hat{y}_t = y_{t-1}$ (Hewamalage, Ackermann, & Bergmeir, 2023). The seasonal naive variant generalises this idea to seasonal data by predicting the observation from the corresponding point in the previous season, $\hat{y}_t = y_{t-m}$, where m denotes the seasonal period (e.g., $m = 7$ for daily data with a weekly pattern).

The Autoregressive Integrated Moving Average model, $\text{ARIMA}(p, d, q)$, regresses y'_t on its p most recent lagged values, applies d -th order differencing for stationarity, and regresses on the previous q forecast errors:

$$y'_t = c + \sum_{i=1}^p \phi_i y'_{t-i} + \sum_{i=1}^q \theta_i \varepsilon_{t-i} + \varepsilon_t \quad (7)$$

with y'_t the d -times differenced series and ε_t white noise. Orders are selected by minimising AICc, with d determined via unit-root tests (Hyndman et al., 2025).

5.1 Machine Learning Methods

Gradient Boosted Decision Trees (GBDT) (Friedman, 2001) construct a predictor as a forward stage-wise additive expansion of regression trees, each fitted to the negative gradient of the loss at the current ensemble's predictions. Their native handling of missing values, mixed feature types, and non-linear interactions has made them a dominant choice for tabular prediction (So, 2024).

XGBoost, LightGBM and CatBoost dominate applied gradient boosting. All share the additive tree-boosting core but differ in ways that matter for a sparse, over-dispersed target (So, 2024): XGBoost favours regularised, conservative tree growth; LightGBM trains fastest and leads tabular time-series benchmarks (Hall & Rasheed, 2025; Makridakis et al., 2022) but is more prone to overfitting on limited data; and CatBoost's ordered boosting tends to be more robust on smaller, noisier samples. Because their relative ranking is data-dependent, all three are evaluated rather than chosen a priori.

Among GBDT implementations, **XGBoost** (T. Chen & Guestrin, 2016) scales gradient boosting to large tabular problems through two contributions: a regularised objective and a second-order optimisation step. For a dataset with n examples, the model predicts via K additive CART functions:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F}, \quad (8)$$

where $\mathcal{F}$ is the space of regression trees. The training objective is

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k), \quad \text{with} \quad \Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2, \quad (9)$$

in which T is the number of leaves and w the leaf weights. The penalty Ω smooths the learnt weights and prefers simpler trees, mitigating overfitting

(T. Chen & Guestrin, 2016). Because $\mathcal{L}$ cannot be optimised in functional space directly, XGBoost expands it to second order around the prediction at iteration $t - 1$, giving a closed-form leaf score and a tractable split-gain criterion based on the first- and second-order gradient statistics of the loss. Practically, XGBoost grows trees *level-wise* up to a maximum depth, then prunes splits with non-positive gain (So, 2024).

LightGBM (Ke et al., 2017) retains XGBoost’s regularised objective but introduces Gradient-based One-Side Sampling (GOSS) — retaining all large-gradient instances while sub-sampling small-gradient ones — and Exclusive Feature Bundling (EFB), which merges mutually exclusive sparse features. Trees grow *leaf-wise*, splitting the leaf with the largest loss reduction, which reaches lower training loss in fewer iterations at the cost of overfitting risk on smaller datasets (Ke et al., 2017; So, 2024).

CatBoost (Prokhorenkova, Gusev, Vorobev, Dorogush, & Gulin, 2018) introduces ordered target statistics and ordered boosting to eliminate target leakage and prediction shift. Both techniques rely on a random permutation of the dataset: categorical features are encoded using expectations from preceding examples only, and gradients are computed using the same permutation sequence. Additionally, CatBoost utilizes oblivious decision trees as base learners. By enforcing identical split conditions at every node of a given depth, these models produce balanced trees optimized for fast inference (So, 2024).

5.2 Deep Learning Methods

Deep learning methods are a serious contender in time series forecasting, particularly when relationships among lagged covariates are non-linear or when many parallel series share structure (Lim & Zohren, 2021; Mendis et al., 2024). Two families are examined — Recurrent Neural Networks (RNNs) and a pretrained foundation model — selected for different data demands and efficiencies, from the higher-capacity LSTM, to the lighter GRU, to the pretrained Chronos-2. Unlike the tree-based learners, all three can in principle extrapolate beyond the counts seen in training, motivating their evaluation against the gradient-boosted trees.

RNNs maintain a hidden state $h_t = \sigma(W_h h_{t-1} + W_x x_t + b)$ updated at each step (Elman, 1990). The recurrent product of weight matrices causes backpropagated gradients to vanish or explode over long horizons, which the gated LSTM and GRU variants were designed to address (Hochreiter & Schmidhuber, 1997).

The **Long Short-Term Memory** (LSTM) (Hochreiter & Schmidhuber, 1997) augments the recurrent unit with a separate cell state c_t that flows through the sequence with only linear, gated updates, preserving gradient

magnitude across many time steps. At each step, three sigmoid gates—forget, input, and output—control information flow:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f), \quad (10)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i), \quad (11)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o), \quad (12)$$

The cell and hidden states are updated as

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c), \quad (13)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad (14)$$

$$h_t = o_t \odot \tanh(c_t), \quad (15)$$

where $\odot$ denotes element-wise multiplication.

The **Gated Recurrent Unit** (GRU) (Cho et al., 2014) drops the separate cell state and uses two gates — a reset gate r_t and an update gate z_t — making it faster to train:

$$r_t, z_t = \sigma(W_{r,z}[h_{t-1}, x_t] + b_{r,z}), \quad (16)$$

$$\tilde{h}_t = \tanh(W_h[r_t \odot h_{t-1}, x_t] + b_h), \quad (17)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t. \quad (18)$$

In practice, training restricts effective memory to a fixed lookback window (the input chunk length). Both LSTM and GRU are trained under Poisson, Tweedie, and MSE objectives (Section 5.3).

As for the foundation model, Chronos-2 (Ansari et al., 2025) is a pre-trained encoder-only transformer for time-series forecasting, applied zero-shot or fine-tuned to a downstream task, that transfers structure acquired from a large mixed synthetic/real pretraining corpus.

Input channels are robust-scaled, $\sinh^{-1}$ -transformed, and split into patches. Transformer blocks alternate *time-attention* (along the temporal axis within a channel) with *group-attention* (across channels sharing a group identifier). The latter binds the target to its past-only and known-future covariates, allowing information to flow among them. A residual head projects the output onto 21 quantile levels under the pinball loss; the median is taken as the point forecast for comparability.

The model is invoked through the `autogluon-timeseries` wrapper. Each region is treated as one item with a seven-day prediction length, so that a single call returns a forecast at every lead time from one to seven days ahead; weather and calendar fields enter as *known* covariates (future values observable at the forecast origin), the remaining engineered features

as *past-only* covariates, and the per-region activity tier as the single *static* covariate. Two configurations are evaluated: a *zero-shot* predictor using the pretrained weights unchanged, and a *fine-tuned* variant.

5.3 Loss functions

All count-style models in this thesis (the LightGBM, XGBoost, and CatBoost gradient boosters as well as the LSTM) are trained with a log-link parametrisation: the raw model output p is mapped to a strictly positive predicted mean via $\hat{y} = e^p$. This is the canonical link for both the Poisson and Tweedie families (Qian, Yang, & Zou, 2016; Zhou et al., 2022), and ensures that predicted means remain non-negative regardless of the sign of p . In the GBDTs the log-link is applied by the library objective; in the LSTM it is imposed through the custom loss modules below. A baseline LSTM is additionally trained with Mean Squared Error on the raw target scale, without log-link.

In the following, $y \geq 0$ denotes the observed strike count and $\hat{y}$ the predicted mean: $\hat{y} = e^p$ for the count models, and a point prediction on the target scale for the Mean Squared Error (MSE) baseline.

MSE: $\mathcal{L}_{\text{MSE}}(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$, is the standard regression loss. It treats the target as approximately Gaussian with constant variance, a poor match for non-negative, zero-inflated count data; here it serves only the regression task on the first-differenced target series.

For discrete count regression the Poisson negative log-likelihood (NLL) is the standard objective. With respect to the mean it reads

$$\mathcal{L}(y, \hat{y}) = \hat{y} - y \log \hat{y} + \log(y!), \quad (19)$$

where the $\log(y!)$ term is constant in $\hat{y}$ and therefore contributes nothing to the gradient. Dropping it and substituting the log-link $\hat{y} = e^p$ yields the loss that is actually minimised during training:

$$\mathcal{L}_{\text{Poi}}(y, p) = e^p - y \cdot p. \quad (20)$$

The Tweedie family is a sub-class of exponential dispersion models with variance function $\text{Var}(Y) = \phi \mu^\rho$, parameterised by a power ρ . For $\rho \in (1, 2)$ the corresponding distribution is the compound Poisson–Gamma, which assigns positive probability mass to $y = 0$ and a continuous, right-skewed density on $y > 0$ (Qian et al., 2016; Zhou et al., 2022). This is precisely the structure expected of zero-inflated, non-negative targets such as daily strike counts per region.

Up to additive constants in $\hat{y}$, the Tweedie NLL with respect to the mean is

$$\mathcal{L}(y, \hat{y}) = -y \frac{\hat{y}^{1-\rho}}{1-\rho} + \frac{\hat{y}^{2-\rho}}{2-\rho}. \quad (21)$$

Substituting $\hat{y} = e^p$ gives the loss optimised in practice:

$$\mathcal{L}_{\text{Tw}}(y, p) = -y \frac{e^{(1-\rho)p}}{1-\rho} + \frac{e^{(2-\rho)p}}{2-\rho}. \quad (22)$$

The limits $\rho \rightarrow 1$ and $\rho \rightarrow 2$ recover the Poisson and Gamma deviance losses respectively (Zhou et al., 2022).

5.4 Hurdle Model

The hurdle model addresses zero-inflation by decomposing the conditional distribution of an observed event count into two components (Mullahy, 1986; Xu et al., 2024). Let $Y \in \mathbb{N}_0$ denote this observed event count and let X represent the feature vector. A binary model estimates the probability of crossing the hurdle, i.e. of observing at least one event,

$$\hat{\pi}(X) = P(Y > 0 \mid X), \quad (23)$$

while a zero-truncated count model estimates the expected magnitude conditional on occurrence,

$$\hat{\mu}(X) = E[Y \mid X, Y > 0]. \quad (24)$$

By the law of total expectation, and because $E[Y \mid X, Y = 0] = 0$, the unconditional predicted mean is the product of the two components:

$$\hat{E}[Y \mid X] = \hat{\pi}(X) \cdot \hat{\mu}(X). \quad (25)$$

Because the two stages are estimated on disjoint partitions — the classifier on all observations, the count model on the strictly positive subset — their likelihoods factorise and the components can be fitted independently (Mullahy, 1986; Xu et al., 2024).

The binary and count stages are instantiated from the model families described in Sections 5.4 (SPE)–5.1 (Catboost, Tweedie). The final 7-day forecast for each region is obtained by applying Eq. (25) elementwise across the forecast horizon, as illustrated in Figure 9.

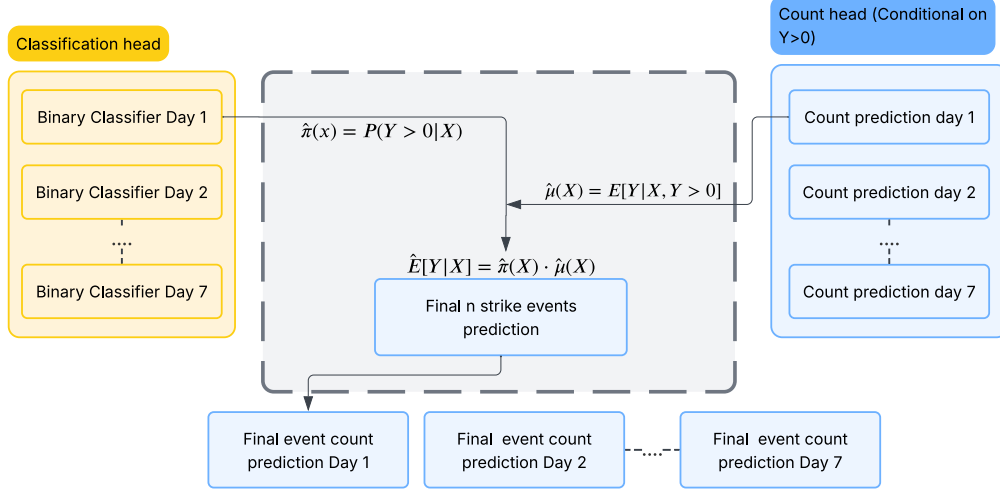


Figure 9: Two-stage hurdle architecture. A classification head produces the daily hurdle probability $\hat{\pi}(X) = P(Y > 0 | X)$ for each of the seven forecast days, and an independently trained count head produces the conditional expectation $\hat{\mu}(X) = E[Y | X, Y > 0]$. The two are combined multiplicatively to yield the unconditional event-count prediction $\hat{E}[Y | X]$ per day.

The hurdle classifier inherits the severe class imbalance of the underlying event data, motivating a dedicated imbalanced-learning method. **The Self-Paced Ensemble (SPE)** (Liu et al., 2020) addresses this regime through an iterative under-sampling scheme that conditions the sampling distribution on the current ensemble’s classification difficulty, rather than on the raw class ratio.

Let $D = P \cup N$ denote the training set, partitioned into a positive set $P = \{(x, y) \mid y = 1\}$ and a negative set $N = \{(x, y) \mid y = 0\}$, with $|P| \ll |N|$. Given a trained binary classifier F returning $F(x) \in [0, 1]$, the per-sample classification hardness is the absolute error

$$\mathcal{H}(x, y, F) = |F(x) - y|, \quad (26)$$

which separates trivial (low- $\mathcal{H}$), borderline, and noisy or mislabelled (high- $\mathcal{H}$) examples. At iteration i , the negative set is partitioned into k equal-width bins $B_1, \dots, B_k$ over $\mathcal{H}$, and the mean hardness h_l is computed per bin. The unnormalised sampling weight of bin B_l is

$$p_l = \frac{1}{h_l + \alpha_i}, \quad \alpha_i = \tan\left(\frac{i\pi}{2n}\right), \quad (27)$$

so that the self-paced factor α_i grows from 0 to ∞ over the n rounds. In early rounds, α_i is small and weights are dominated by $1/h_l$, focusing sampling on informative borderline negatives; in later rounds, α_i dominates and weights converge towards uniformity, retaining easy samples as a

stabilising “skeleton” that mitigates overfitting to noisy points. Each round draws $|P|$ negatives proportional to p_l , fits a base learner on the resulting balanced subset, and adds it to the ensemble; the final prediction is the mean over base learners.

A probabilistic classifier is *calibrated* when, among instances assigned predicted probability $\hat{p}$, the empirical positive rate is approximately $\hat{p}$ (Kull, Silva Filho, & Flach, 2017). In the hurdle architecture, $\hat{\pi}(X)$ is not thresholded but enters multiplicatively into Equation (25), so any bias propagates directly into $\hat{E}[Y | X]$ — making calibration a precondition for unbiased predictions.

The under-sampling at SPE’s core further alters the class prior seen by each base learner, shifting probability estimates away from the population prior and motivating a post-hoc calibration map fit on training data (Kull et al., 2017).

Platt scaling (Platt et al., 1999) fits a logistic sigmoid to raw scores under a Gaussian-equal-variance assumption that fails for the bounded, skewed scores of imbalanced classifiers, sometimes worsening calibration. Kull et al. (2017) instead assume per-class scores follow Beta distributions on $[0, 1]$, giving a three-parameter family that contains the identity map; restricting the shape parameters to $a = b$ yields a two-parameter sub-family that keeps this identity property and is preferred here for sample efficiency, as the full calibrator is liable to overfit on modest validation folds. By Proposition 1 of Kull et al. (2017), fitting Beta[$a = b$] is equivalent to a univariate logistic regression on log-odds-transformed scores.

In practice, the raw SPE probabilities are clipped, logit-transformed, and passed through a univariate logistic regression fit by maximum likelihood, following Algorithm 1 of Kull et al. (2017, p. 5063); test-set probabilities are passed through the fitted regressor.

6 EXPERIMENTAL SETUP

Figure 10 provides a schematic overview of the experimental pipeline. Target data (ACLED drone-strike events) and exogenous sources are independently cleaned, filtered, and engineered into a common pool of region-day series, partitioned chronologically into training, validation, and test sets (Section 4.2); model-specific feature engineering and selection (Section 4.5) are then applied per branch. Four model families—classical baselines, gradient-boosted trees, deep sequential networks, and the hurdle’s Self-Paced Ensemble—are tuned via expanding-window backtests on the validation set and evaluated on the held-out test set, with feature-importance diagnostics extracted from the best configurations (Section 7).

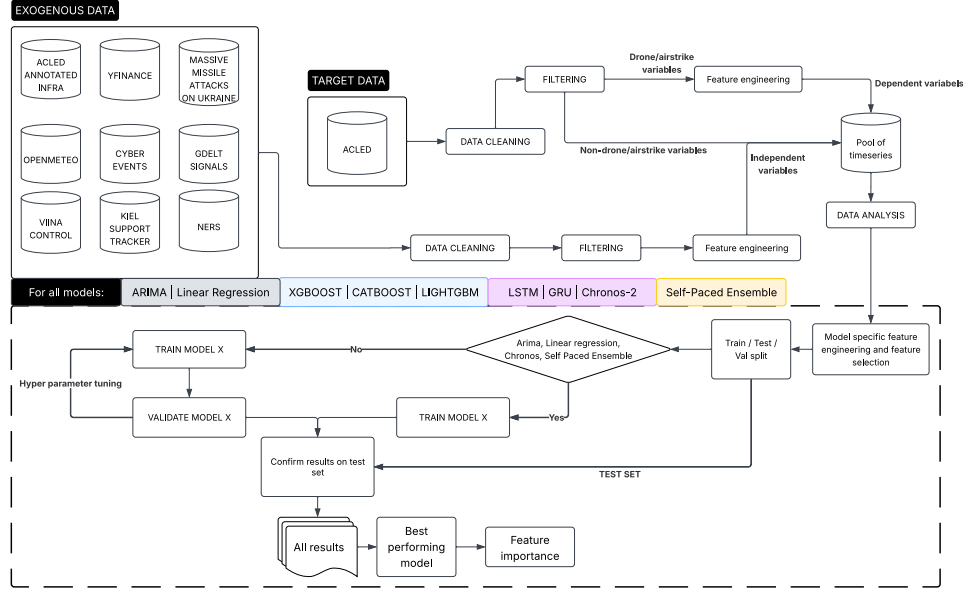


Figure 10: Overview of the experimental pipeline.

6.1 Backtesting protocol

Model performance is assessed using a rolling-origin protocol, the time-series analogue of k -fold cross-validation (Hewamalage et al., 2023): rather than randomising observations into folds, it preserves temporal ordering and advances the forecast origin through the series, reproducing the conditions under which a deployed model would operate. Of the two principal variants (Hewamalage et al., 2023), the *expanding window* is adopted—the training set is monotonically enlarged rather than translated forward at fixed length—motivated by the moderate series length, for which retaining all history is preferable to discarding early data.

Under the chronological 70/10/20 split of Section 4.2, the validation set serves two purposes: hyperparameter tuning and fitting the Beta[$a=b$] calibrator of Section 5.4. The test set is withheld until final evaluation. Within the test segment the forecast origin advances one day at a time: at each day d , the current model issues forecasts for all seven lead times $d+1, \dots, d+7$ simultaneously.

The RNN based models inherently handle the multiple outputs by number of neurons in the output layer, for chronos-2 this is handled by the multi-patch quantile outputs. For the Gradient boosted decision tree based models, however, the multi-horizon forecasts are produced under a *direct* strategy. Let d denote the current forecast origin and let $h \in \{1, \dots, H\}$ with $H=7$ index the forecast horizon, so that $d+h$ is the h th target day. A

separate model $f_h(\cdot)$ is trained for each horizon, yielding $\hat{y}_{d+h} = f_h(\mathbf{x}_d)$ from the covariate vector $\mathbf{x}_d$ available at the origin. Unlike the recursive strategy, which iterates a one-step model and accumulates error across horizons, the direct strategy avoids this—capturing horizon-specific feature relevance—at the cost of training H models per configuration.

Following the distinction drawn by Hewamalage et al. (2023) between *updating* (feeding new observations to a model without retraining) and *recalibrating* (refitting on the extended history), the model is recalibrated only after seven new observations have accumulated, at which point each of the H horizon-specific models is refit on the extended training period $\mathcal{P}+7$.

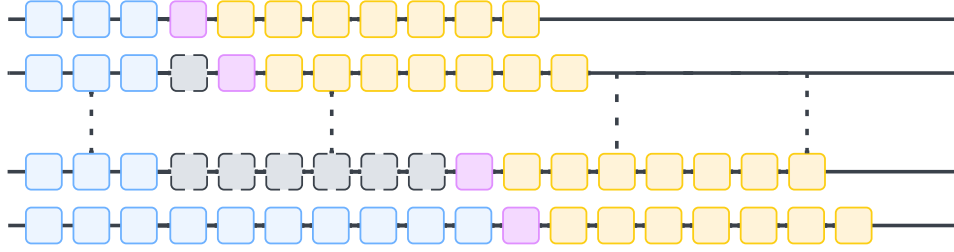


Figure 11: Expanding-window evaluation with daily forecast origins and weekly recalibration. At each day d the model fit on training period $\mathcal{P}$ issues forecasts for $d+1, \dots, d+7$; after seven successive advances of the origin the model is recalibrated on the extended training period $\mathcal{P}+7$ and the cycle repeats. **Blue:** observations used to fit the current model. **Purple:** the current forecast origin d . **Yellow:** the seven forecast targets $d+1, \dots, d+7$. **Grey (dashed):** observations that have been forecast at a previous origin but are not yet incorporated into the training set, awaiting the next recalibration.

Hyperparameters were tuned by Bayesian optimisation, specifically the Tree-structured Parzen Estimator (TPE) (Bergstra, Bardenet, Bengio, & Kégl, 2011) as implemented in Optuna (Akiba, Sano, Yanase, Ohta, & Koyama, 2019). TPE models $p(x | y)$ by splitting evaluated configurations at a loss quantile y^* into densities $\ell(x)$ (good) and $g(x)$ (bad), then selects candidates maximising the ratio $\ell(x)/g(x)$, equivalent to Expected Improvement.

6.2 Evaluation metrics

Model performance is evaluated with unscaled point-forecast errors (MAE, RMSE), a scale-free measure (RMSSE) for cross-region aggregation, and a threshold-free classification metric (PR-AUC) for the hurdle’s binary classification head.

Two unscaled errors are reported on the original count scale: RMSE (Eq. 28) penalises deviations quadratically, MAE (Eq. 29) linearly. Although common in regression, R^2 is omitted because it can be misleading in time-series forecasting, especially for true random-walk series. (Hewamalage et al., 2023).

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (28)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (29)$$

RMSE is reported as the primary metric: misses on high-intensity days are operationally costly, the quadratic penalty reflects this asymmetry. MAE is reported as a secondary metric for typical-day performance in the predominantly zero-inflated regions.

Skill score introduced as additional metric. Skill score, adapted from (Roebber, 1998), shows the relative performance improvement over the naive seasonal baseline

$$SS_{\text{model}} = 1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{baseline}}} \quad (30)$$

The hurdle classifier is evaluated using PR-AUC, which integrates precision against recall across thresholds. Unlike ROC-AUC — which factors in the large true-negative class and remains optimistic under severe imbalance — PR-AUC isolates minority-class performance by penalising false positives directly (Saito & Rehmsmeier, 2015).

Following standard statistical procedures as described in Navarro (2015), the normality of the distributions was assessed using the Shapiro-Wilk test, and the homogeneity of variances with Levene’s test. Where both parametric assumptions held, a one-way analysis of variance (ANOVA) was conducted with Tukey’s Honestly Significant Difference (HSD) post-hoc test for pairwise comparisons; otherwise the non-parametric Kruskal-Wallis H-test was used to assess differences in medians.

For post-hoc analysis of statistical significance in model performance, a blocked, rank-based comparison was conducted following Demšar (2006). Each (region, horizon) cell forms one block ($N = 140$); within every block the k competing models are ranked by error. Following the same assumption check used for the horizon analysis, the rank-based Friedman test is applied where the blocked-ANOVA normality assumption is not met. The Friedman test evaluates equality of mean ranks, and where it is rejected the Nemenyi post-hoc test flags pairs differing by more than the

critical distance $CD = q_\alpha \sqrt{k(k+1)/6N}$. The critical value q_α is derived from the Studentized range statistic divided by $\sqrt{2}$ (see Table 5 in [Demšar, 2006](#))

6.3 Feature importance

For the GBDT models, two complementary importance measures are reported: the native gain-based importance and a post-hoc permutation importance.

Gain-based importance, computed internally by XGBoost, LightGBM, and CatBoost, attributes to each feature j the total reduction in the training objective across all splits in which it is selected,

$$I_{\text{gain}}(j) = \sum_{k=1}^K \sum_{s \in S_k^{(j)}} \Delta \mathcal{L}_s, \quad (31)$$

where $S_k^{(j)}$ collects the splits in tree k that use feature j and $\Delta \mathcal{L}_s$ is the loss reduction at split s . It is essentially free to obtain but is a structural quantity: it reflects how the trained ensemble *uses* a feature, not how much out-of-sample accuracy depends on it.

Permutation importance ([Strobl, Boulesteix, Zeileis, & Hothorn, 2007](#)) is defined on held-out data. Letting $\mathcal{L}$ denote the validation loss and $\pi_j(X)$ the design matrix with column j randomly shuffled, the importance of j is the resulting loss increase,

$$I_{\text{perm}}(j) = \mathcal{L}(f(\pi_j(X)), y) - \mathcal{L}(f(X), y), \quad (32)$$

averaged over repeated permutations. It is model-agnostic and measures the marginal contribution of j to predictive performance on the evaluation set.

The two are reported jointly because their biases differ: gain-based importance favours high-cardinality features even when their marginal value is modest ([Strobl et al., 2007](#)), whereas permutation importance avoids this but understates information shared across correlated covariates. Their disagreement is therefore diagnostic—concordance marks a robustly important feature—and combining them reduces estimate variance ([Rengasamy, Rothwell, & Figueredo, 2021](#)).

7 RESULTS

To evaluate the predictive capabilities of the selected algorithms, a large set of model configurations were trained and hyperparameter-tuned on

the validation set. The gradient boosted decision tree models (CatBoost, LightGBM, XGBoost) were evaluated under three training paradigms: Global (trained on all data simultaneously), Activity (trained exclusively on data from regions sharing the same activity tier), and Local (an independent model trained per region) under two count-distributional loss functions (Poisson and Tweedie). The deep learning models (LSTM and GRU) were evaluated under the Global and Activity paradigms and tuned for input windows of 7, 14, and 28 days using both Poisson and Tweedie loss functions; Local evaluation was omitted due to insufficient data volume per region to train deep sequential models. Both the GBDT models and RNN based models were additionally evaluated with the MSE loss function on the differenced target variable under the same Global, Activity and Local paradigms.

Chronos-2 was evaluated in both zero-shot and fine-tuned capacities, treating the regions locally as a multivariate time series; ARIMA was fitted strictly as a local model, since it cannot natively ingest exogenous variables. All methods were implemented in Python², primarily using the Darts library for time-series modelling (Herzen et al., 2022) and AutoGluon to access Chronos-2 (Shchur et al., 2023).

All hyperparameters were optimised with the TPE sampler of Optuna (Akiba et al., 2019) under a MedianPruner ($n_{\text{warmup}} = 5$), seed 42, using a three-fold expanding-window backtest. Each Poisson, Tweedie, and (where applicable) MSE-on-differenced-target variant received its own study; for the recurrent models, the input chunk lengths $w \in \{7, 14, 28\}$ were likewise tuned as separate studies. Each GBDT and RNN study ran for 50 trials. The Chronos-2 fine-tuning study, with substantially higher per-trial cost, ran for 12 trials. Tuning was conducted under the *Global* paradigm; the Activity- and Local-paradigm runs reuse the Global-tuned configuration. Search spaces and best configurations for the top-five models in Table 5 are reported together with hardware specifications in Appendix C.

The remaining results are organised around the sub-questions: Section 7.1 compares the machine- and deep-learning families and evaluates the hurdle model as a separate contender (Sub-Q1); Section 7.2 characterises performance across the one-to-seven-day horizon (Sub-Q2); and Section 7.3 identifies the most informative predictors for the top models (Sub-Q3).

² Python Software Foundation. Python Language Reference, version 3.15. Available at <https://www.python.org>

7.1 Overall Model Performance

Table 4 reports the best-performing configuration of each model family across the test window. The primary metric is the Skill Score, which expresses the fractional improvement in RMSE relative to the seasonal-naive baseline. Table 5 restricts this to the top five.

Model	Paradigm	Configuration	MAE	RMSE	Skill Score
CatBoost	Activity	Tweedie	0.77	1.83	0.17
LightGBM	Global	Poisson	0.78	1.84	0.17
XGBoost	Global	Tweedie	0.78	1.87	0.15
LSTM	Activity	Poisson, $w = 28$	0.78	1.84	0.17
GRU	Activity	Poisson, $w = 28$	0.78	1.87	0.15
Chronos-2-FT	Local	Fine-tuned	0.71	1.83	0.17
Chronos-2-OS	Local	Zero-shot	0.72	1.85	0.16
ARIMA	Local	$p = 7, d = 1, q = 1$	0.81	1.85	0.16
Hurdle	Global	SPE + CatBoost (Tweedie)	0.79	2.00	0.09
Seasonal Naive	Local	$m = 7$	0.89	2.20	0.00
Naive	Local	-	1.03	2.58	-0.17

Table 4: Best performing paradigm and configuration per model type. Skill Score represents the fractional improvement over the Seasonal Naive baseline.

Table 5: Ranked list of top 5 performing models overall, ordered by Skill Score.

Rank	Paradigm	Model	Skill Score	MAE	RMSE
1	Activity	CatBoost (Tweedie)	0.170	0.78	1.83
2	Local	Chronos-2 (Fine-tuned)	0.169	0.71	1.83
3	Global	CatBoost (Tweedie)	0.169	0.79	1.83
4	Global	LightGBM (Poisson)	0.166	0.78	1.84
5	Activity	LSTM (Poisson, $w = 28$)	0.165	0.78	1.84

The top configurations cluster within a narrow band (RMSE 1.83 - 1.87, SkillScore 0.15-0.17) spanning CatBoost, LightGBM, LSTM, GRU and Chronos-2. Activity CatBoost-Tweedie ranks highest (Skill Score = 0.170), closely followed by the fine-tuned Chronos-2 and the Global CatBoost-Tweedie variant (Skill Score = 0.169 each). The fine-tuned Chronos-2 attains the lowest MAE (0.71), exceeding the next-best model by approximately 0.06. Despite accessing only univariate target history, ARIMA lands

within 0.02 RMSE of the leaders and ahead of several gradient-boosted and recurrent variants (Table 4; full ranking in Appendix E).

Despite the increase of stationarity, no differenced-target plus MSE loss configuration reaches the leading group; the best (Global CatBoost) attains a Skill Score of only 0.06. Per-region, the top five beat the seasonal-naive baseline under RMSE in all twenty regions (Figure 12). Under MAE the baseline is more competitive: in Zaporizhia, Odesa and Lviv it outranks every leading model but the local Chronos-2 (Figure 13).

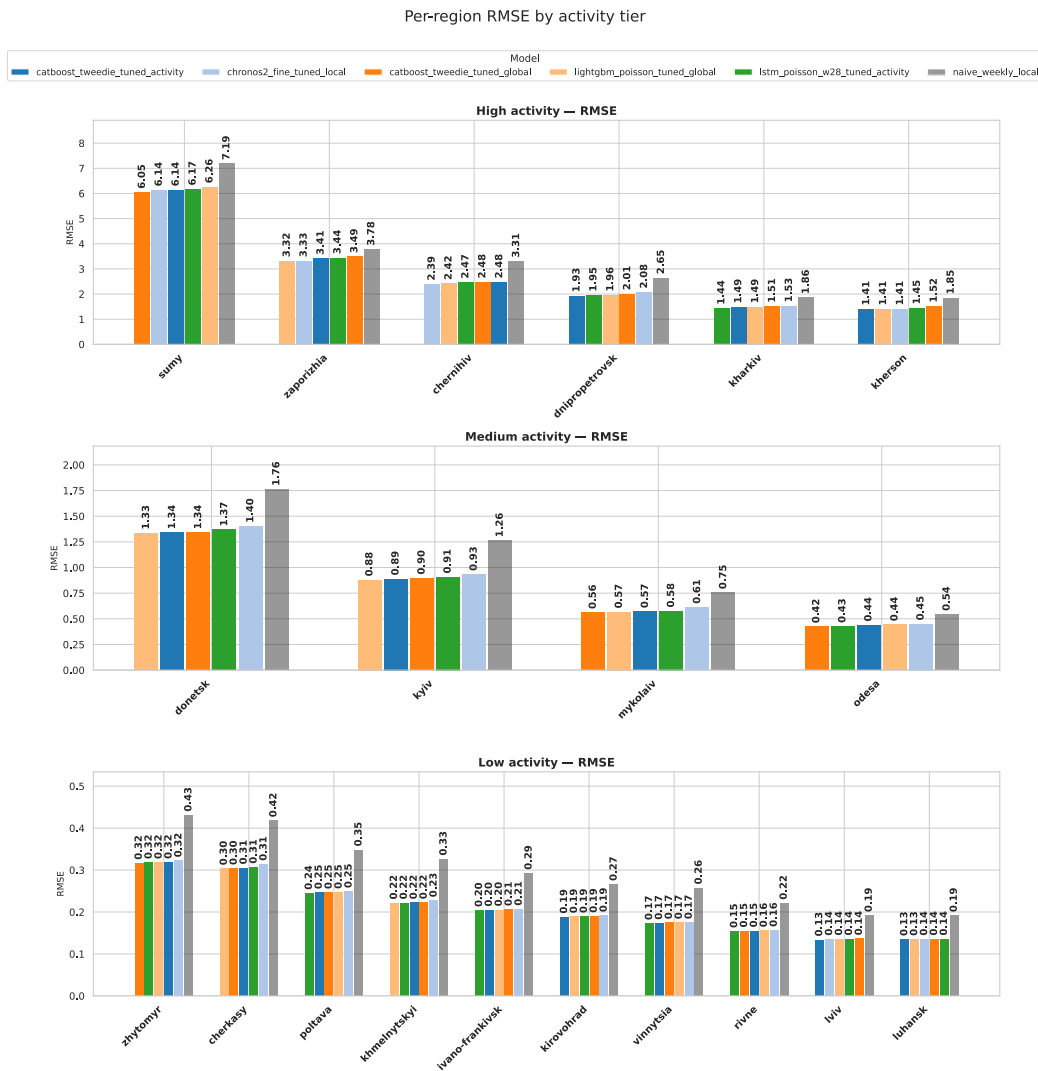


Figure 12: Per-region test RMSE for the five leading models and the seasonal-naive baseline, grouped by activity tier (note the independent y-axis per panel).

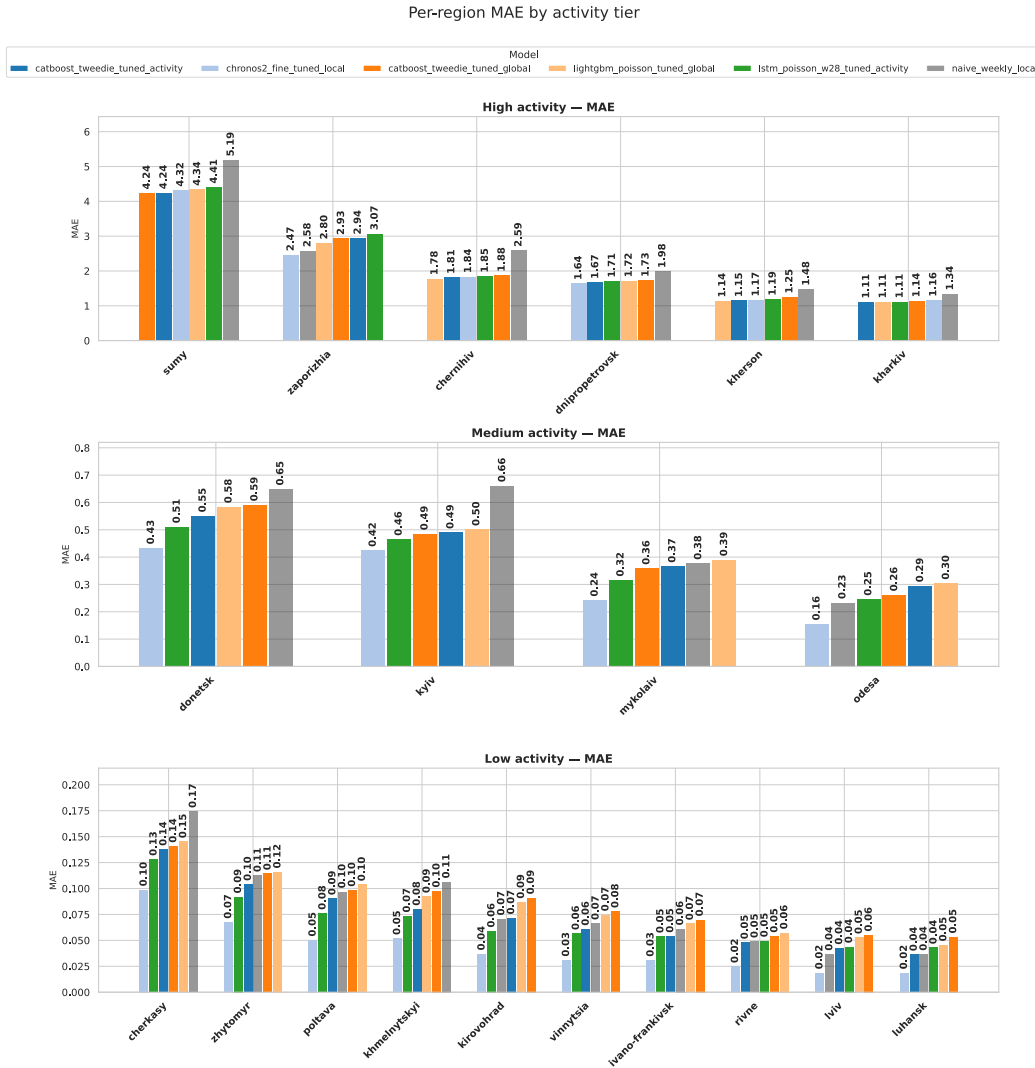


Figure 13: Per-region test MAE for the five leading models and the seasonal-naive baseline, grouped by activity tier (note the independent y-axis per panel).

The narrow spread in Table 5 raises the question of whether the top models are statistically distinguishable. The blocked-ANOVA residuals are non-normal for both metrics RMSE and MAE (Shapiro–Wilk, $p < 0.001$). A blocked Friedman test ($\alpha = 0.05$) over the 140 (region, horizon) cells is applied to the top-five configurations plus ARIMA. It rejects equal mean ranks for both (RMSE $p < 0.001$; MAE $p < 0.001$; $CD = 0.64$). Under RMSE (Figure 14), Activity CatBoost-Tweedie (2.65), Global LightGBM-Poisson (2.75) and Activity LSTM-Poisson (3.14) are mutually indistinguishable, while ARIMA (4.51) and fine-tuned Chronos-2 (4.62) rank significantly below them; so ARIMA is consistently outranked block-wise despite its

near-equal pooled RMSE. Under MAE (Figure 15) Chronos-2 rises to the top (1.59) and is significantly best, with Activity LSTM-Poisson (2.89), Activity CatBoost-Tweedie (2.91) next and ARIMA last(4.81). No model configuration performs best on both metrics.

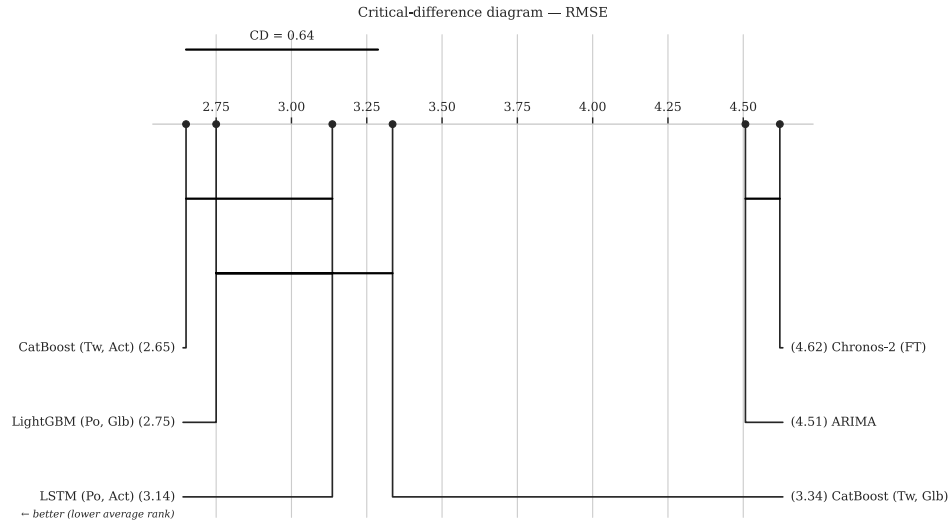


Figure 14: Critical-difference diagram for RMSE over the top-five configurations plus ARIMA, from a Nemenyi post-hoc test on the blocked Friedman ranks across the 140 (region, horizon) cells ($\alpha = 0.05$, $CD = 0.64$). Lower average rank (left) is better; models joined by a horizontal bar are not significantly distinguishable.

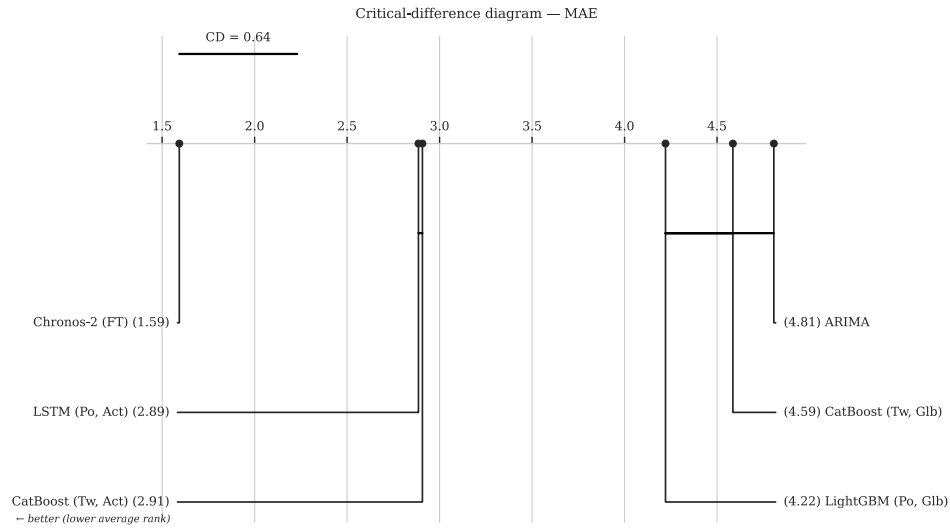


Figure 15: Critical-difference diagram for MAE over the top-five configurations plus ARIMA, from a Nemenyi post-hoc test on the blocked Friedman ranks across the 140 (region, horizon) cells ($\alpha = 0.05$, $CD = 0.64$). Lower average rank (left) is better; models joined by a horizontal bar are not significantly distinguishable.

7.1.1 Hurdle Model Diagnostics

The Hurdle configuration, combining a Self-Paced Ensemble (SPE) classifier with a Beta $[a=b]$ -calibrated probability head and a CatBoost-Tweedie conditional count head, attains a Skill Score of 0.09 (RMSE = 2.00, MAE = 0.79).

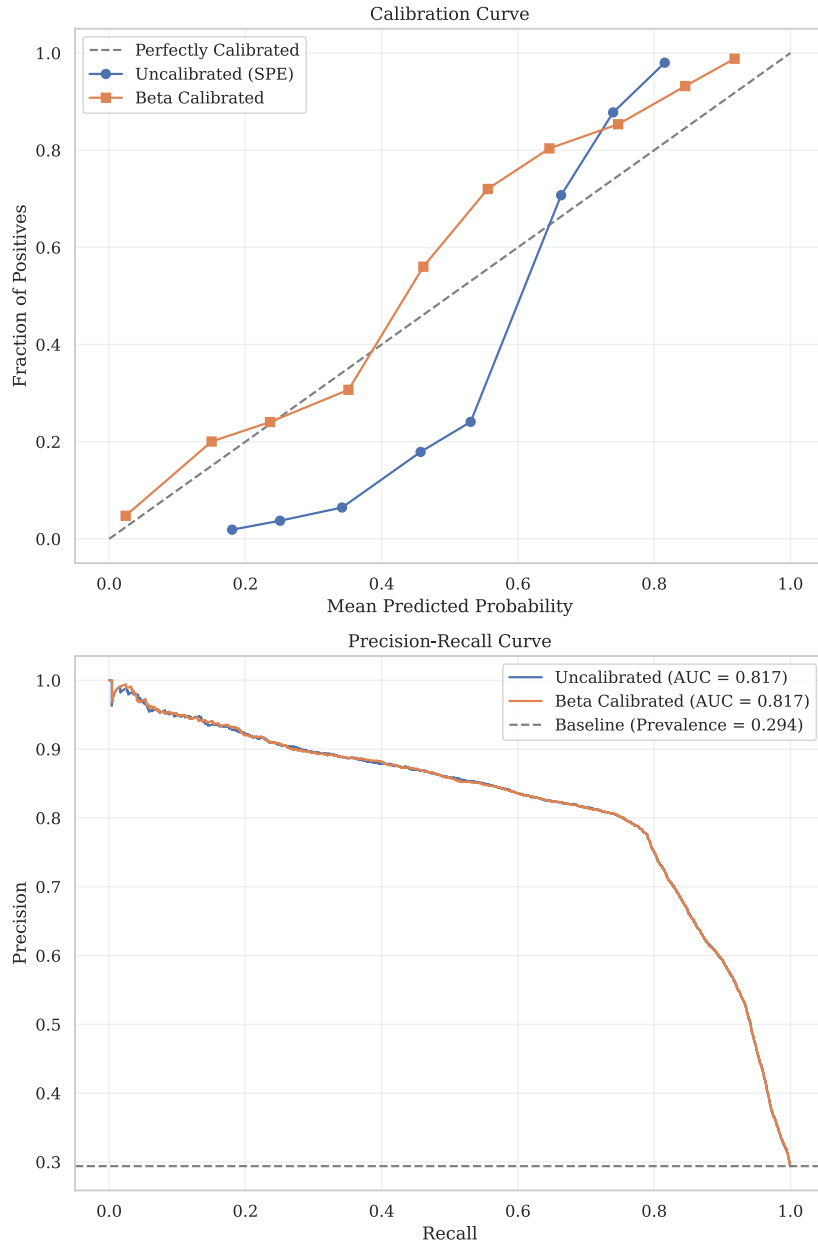


Figure 16: Calibration and precision-recall curves for the uncalibrated and Beta $[a=b]$ -calibrated SPE classifier head. The dashed baseline on the bottom panel marks the positive-class prevalence (0.294).

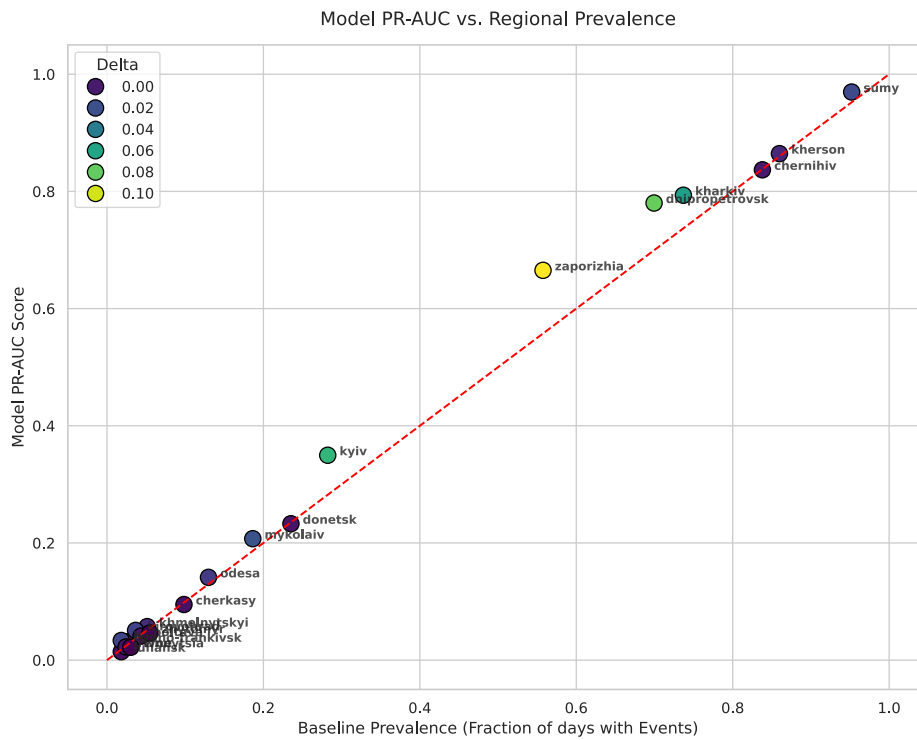


Figure 17: Classifier PR-AUC against regional positive-class prevalence. The dashed line indicates the prevalence baseline; the colour scale encodes the prevalence-adjusted lift $\Delta = \text{PR-AUC} - \text{prevalence}$.

Figure 16 reports the reliability and precision–recall curves for the SPE classifier, both before and after Beta-calibration (Section 5.4).

Two consequences of the calibration step are visible in the figure. First, the reliability curve is pulled close to the diagonal across the full range of predicted probabilities, confirming that the Beta map corrects the under-confidence of SPE at the low-probability range (predicted probabilities of 0.2–0.5 correspond to empirical positive rates near zero) and its over-confidence at the high end. Second, the threshold-free PR-AUC is unchanged (uncalibrated = 0.817, Beta-calibrated = 0.817). This is the expected behaviour: PR-AUC depends only on the ranking of scores, and the Beta[$a=b$] map is strictly monotonic. Following the evaluation framework of Saito and Rehmsmeier (2015), the baseline for the PR curve is dictated by class prevalence. By plotting PR-AUC against regional prevalence (Figure 17), a Delta score is extracted, representing actual predictive gain. The classifier demonstrated hardly any improvement in low-prevalence regions (Tier 1). While PR-AUC scores were naturally higher in high-prevalence areas (Tier 3), the actual improvement over the baseline remained marginal. The most substantial performance gains (Δ between 0.06 and 0.10) were

localized to medium-prevalence regions with incidence rates between 0.28 (Kyiv) and 0.73 (Kharkiv).

Per-region error and bias diagnostics for the conditional count head are reported in Table 6. The mean error is negative in eight evaluated regions, with magnitudes ranging from -0.85 (Kyiv) to -4.17 (Sumy). Bias magnitude scales approximately monotonically with the maximum observed daily strike intensity per region.

Table 6: Hurdle Model Error and Bias Diagnostics per Region

Region	Count (Days > 0)	Max Actual Strikes	RMSE	MAE	Bias (Mean Error)
Sumy	1093	37.0	7.35	4.83	-4.17
Zaporizhia	640	12.0	3.91	3.31	-2.89
Chernihiv	962	11.0	2.70	1.92	-1.60
Donetsk	270	11.0	2.50	1.15	-1.09
Dnipropetrovsk	803	7.0	2.22	1.86	-1.59
Kharkiv	846	8.0	1.57	1.08	-0.91
Kherson	987	5.0	1.46	1.13	-0.87
Kyiv	324	6.0	1.36	0.85	-0.85

7.2 Multi-Horizon Forecasting Performance

Because the models forecast strike events at each lead time from one to seven days ahead, horizon-dependent performance was analysed using the top twenty configurations ordered by Skill Score.

Comparing Model Performance over Horizons

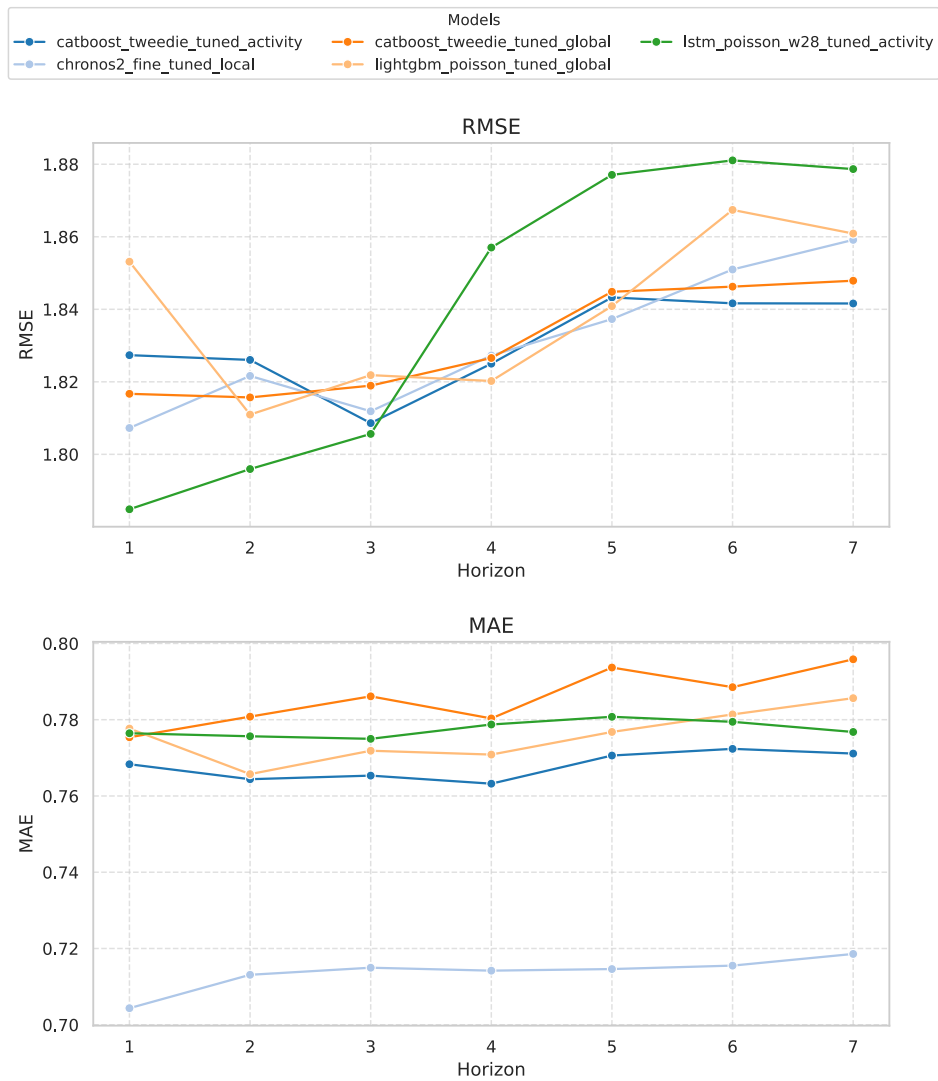


Figure 18: Trajectory of RMSE and MAE across horizons for the top 5 performing models.

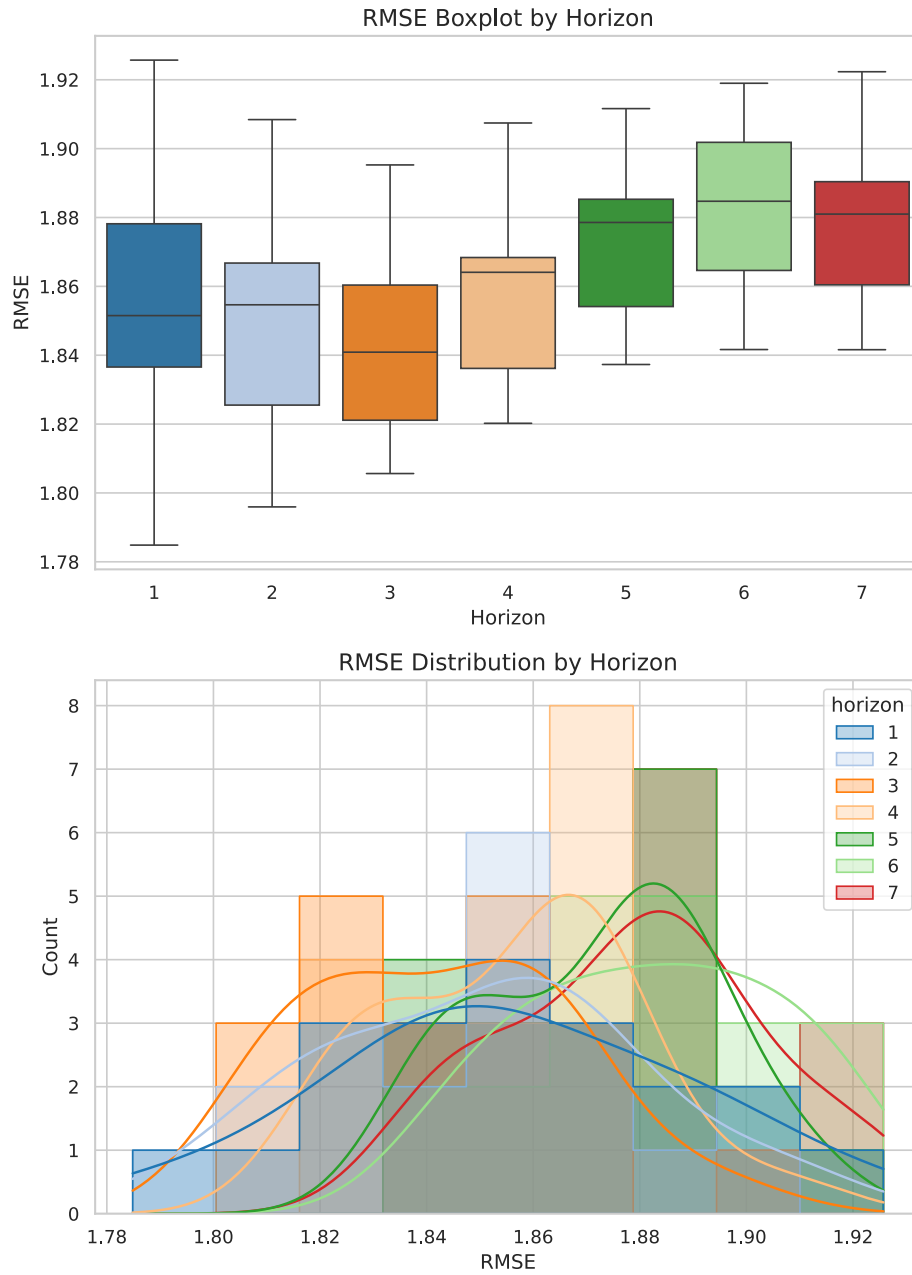


Figure 19: Distribution of RMSE across forecast horizons (Day 1 to Day 7).

The horizon-wise MAE distribution fails the Shapiro-Wilk normality test ($p < 0.05$). A Kruskal-Wallis H-test does not reject the null of equal medians across horizons ($H = 5.49$, $p \approx 0.05$). Across the top five configurations, MAE varies by no more than 0.02 between Day 1 and Day 7

(Figure 18, bottom panel). The fine-tuned Chronos-2 attains the lowest MAE at every horizon, between 0.06 and 0.08 below the next-best model.

The horizon-wise RMSE distribution passes both the Shapiro-Wilk normality test ($p \approx 0.41$) and Levene’s homogeneity-of-variance test ($p \approx 0.54$). A one-way ANOVA rejects equality of means across horizons ($F = 6.70$, $p < 0.001$). Tukey HSD post-hoc comparisons at family-wise $\alpha = 0.05$ identify Day 3 as having the lowest mean RMSE: Day 5 shows a significant increase relative to Day 3 ($p = 0.008$) but not relative to Day 1 ($p = 0.43$), Day 2 ($p = 0.13$) or Day 4 ($p = 0.48$); Day 6 is significantly elevated relative to Days 1 through 4 ($p = 0.027, 0.003, < 0.001, 0.033$ respectively); Day 7 shows the same pattern of significant divergence from each of Days 1–4. Days 1 and 7 are therefore neither the easiest nor the hardest horizons in the ranking.

7.3 Feature importances

Feature importances were extracted from the two best-performing model families: CatBoost-Tweedie under the Activity paradigm (gain-based and permutation; Figure 20) and the fine-tuned Chronos-2 model (permutation only; Figure 21).

To summarise the Activity CatBoost-Tweedie feature importances, each of the top fifteen features is assigned to a semantic category, and every category’s share is computed as its fraction of the summed top-15 importance, separately for the gain and permutation metrics (Figure 20). Each column therefore sums to one. The semantic categories and the full top-15 listings per activity tier are reported in Appendix F.

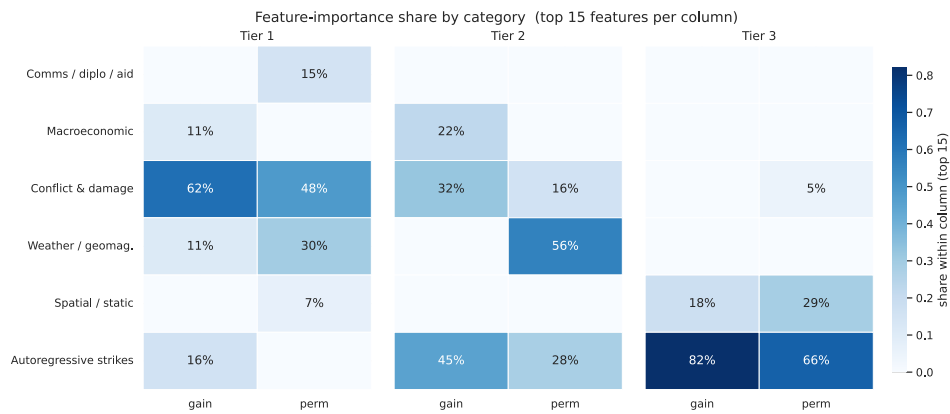


Figure 20: CatBoost-Tweedie feature-importance share by category and activity tier. Each column (information gain vs. permutation) normalises the top-15 features to 100%. Full top-15 listings appear in Appendix F.

Reliance on different feature classes shifts systematically with activity tier, and the two importance metrics agree more closely at the high-activity end than at the low. In tier 1, conflict-and-damage features carry both rankings (62% gain, 48% permutation), but permutation additionally elevates weather and geomagnetic predictors (30%) and communications, diplomacy and aid announcements (15%); Tier 2 is where the metrics diverge most. The gain metric spreads its mass across autoregressive history (45%), conflict-and-damage features (32%) and macroeconomic indicators (22%), whereas permutation is dominated by weather and geomagnetic variables (56%) and relegates the autoregressive and conflict signals to 28% and 16%. In tier 3, both rankings collapse onto two categories: the autoregressive strike history (82% of the gain mass, 66% of the permutation mass) and the static spatial covariate (18% and 29% respectively), with every other category at most marginal.

Chronos2 Local Feature Importance

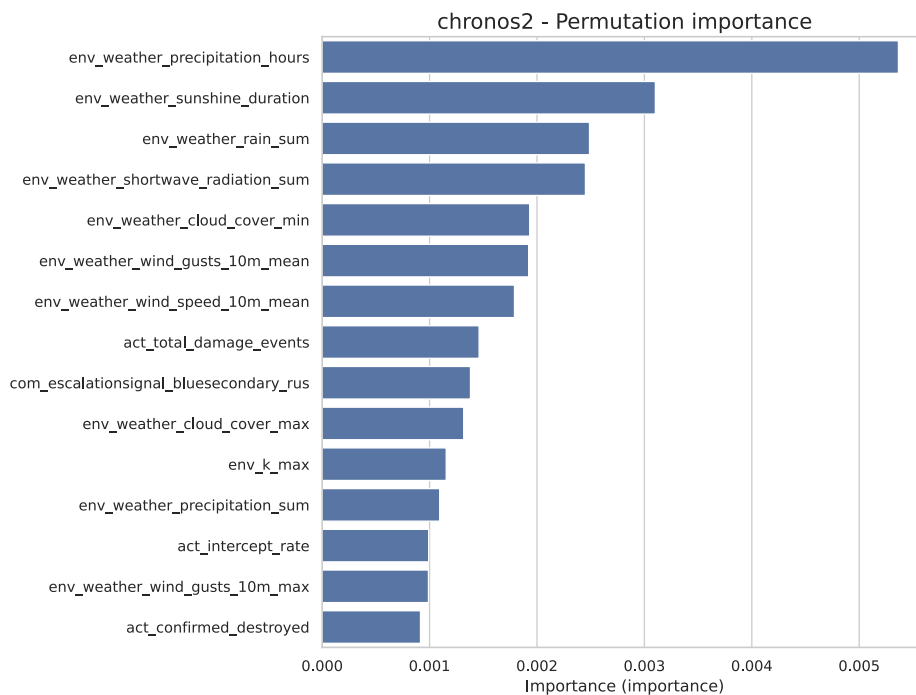


Figure 21: Permutation-based feature importance for the fine-tuned Chronos-2 model.

For the fine-tuned Chronos-2, permutation importance is mostly meteorological: the seven highest-ranked features are all weather variables. Precipitation hours leads, exceeding the second-ranked feature, sunshine duration, by approximately $1.7\times$; rain sum, shortwave-radiation sum, min-

imum cloud cover, and mean 10 m wind gusts and wind speed complete this leading block. The first non-meteorological predictor is the total damage-events count at position eight, followed by the western-Russian secondary-escalation signal at position nine. Unlike the gradient-boosted models, Chronos-2 is not supplied with the engineered lag, rolling-window and exponential-decay covariates.

8 DISCUSSION

This thesis asked to what extent machine-learning and deep-learning algorithms can produce useful multi-horizon spatiotemporal forecasts of Russian-state-initiated drone-strike events across twenty Ukrainian regions, evaluated jointly at every lead time from one to seven days ahead. The research question decomposes into a comparison of model families (Sub-Q1), the behaviour of forecast quality across the horizon (Sub-Q2), and the predictors driving the leading models (Sub-Q3). Section 3 framed this against an empirical gap: the kinetic events of this conflict had been left largely unmodelled; and a methodological gap: these model families had co-occurred in challenges such as VIEWS only as separate submissions, never within one controlled pipeline. The gaps close with asymmetric outcomes: the empirical one constructively, by establishing a reproducible pipeline, dataset, and benchmark; the methodological one analytically, since the controlled comparison shows the families do not meaningfully separate and that the hurdle solution seems mismatched to this data-generating process. Each sub-question is addressed in turn.

For Sub-Q1, the gradient-boosted trees (CatBoost, LightGBM, XGBoost), the recurrent networks (LSTM, GRU), and the fine-tuned Chronos-2 foundation model cluster within a narrow band ($\text{RMSE} \in [1.83, 1.87]$, $\text{Skill Score} \in [0.15, 0.17]$), led by a near-tie on Skill Score (0.17) between Activity CatBoost-Tweedie and the fine-tuned Chronos-2. The two leaders separate only along the choice of metric. On the blocked Friedman ranking across the 140 (region, horizon) cells, Activity CatBoost-Tweedie heads RMSE (rank 2.65, lacking significant separation) while Chronos-2 is significantly best on MAE (rank 1.59) but falls to RMSE rank 4.62; no configuration wins both. This follows from Chronos-2's median-based quantile head, which is conservative on the heavy upper tail: it lowers the typical-day error MAE rewards while leaving the rare high-intensity misses RMSE penalises. The distinction is operational: Chronos-2 where typical-day accuracy matters, the CatBoost (or statistically similar models) where missing a high-intensity day is costlier.

The univariate ARIMA baseline trails the leaders by only 0.02 pooled RMSE, yet ranks significantly lower block-wise (RMSE rank 4.51, MAE 4.81). This shows recent autoregressive history captures most predictable variance, with spatial and exogenous covariates adding a small but statistically notable increment; the feature-importance analysis under Sub-Q3 traces where that increment comes from.

Two patterns hold across model families. First, count-distributional objectives outperform MSE loss on differenced targets (whose best model, Global CatBoost, reaches only a 0.06 Skill Score), showing count likelihood better handles strongly over-dispersed, non-negative data. Second, while LSTM and GRU models matched but did not overtake tree-based models, their performance on these short series suggests a data-limited advantage that could grow with longer histories.

The theoretically motivated hurdle model (an SPE classifier and a CatBoost-Tweedie count head) underperformed every single-stage leader (Skill Score 0.09), for three compounding reasons. The SPE classifier added near-zero lift in low-activity Tier 1 regions, where predicting zero is already strong, and marginal lift in Tier 3, with useful gains confined to medium-prevalence regions. The conditional count head systematically under-predicted high-intensity events, its negative bias scaling with regional intensity (from -0.85 in Kyiv to -4.17 in Sumy). Finally, multiplying the two stages let the upstream probabilities further attenuate an already downward-biased count. Occurrence and magnitude are therefore not cleanly separable here, and a single-stage joint likelihood (e.g. Tweedie) better matches the data-generating process.

For Sub-Q2, the metrics diverge across the horizon. MAE shows no significant variation from Day 1 to Day 7 (Kruskal-Wallis, $p > 0.05$), whereas RMSE degrades significantly beyond the four-day mark (one-way ANOVA $F = 6.70$, $p < 0.001$; Day 5 and later significantly worse than Day 3). Because RMSE penalises large errors most, the deterioration is concentrated in high-intensity outliers while typical-day forecasts stay stable—mirroring the broader pattern that ongoing violence is tracked more reliably than its rare escalations (Bazzi et al., 2022; Hegre et al., 2021). The decay is not strictly monotonic, so Days 1 and 7 are neither easiest nor hardest. One caveat tempers these figures: the models were evaluated on historically accurate weather, whereas a deployed system would use forecasted weather, whose error compounds with lead time and makes the reported multi-horizon performance slightly optimistic.

For Sub-Q3, the gain and permutation rankings of the Activity CatBoost-Tweedie model agree most strongly at high activity. Tier 1 measures partly agree: both rank conflict-and-damage features highest (62% gain, 48% permutation), with weather variables a clear second under permutation (11% gain, 30% permutation). Tier 2 diverges most—gain favours autoregressive strikes (45%) while permutation favours weather (56%), so no single ranking should be over-read in this sparse regime. Tier 3 collapses onto autoregressive history (82% gain, 66% permutation) and the static spatial covariate.

This complements the finding for Sub-Q1: in the high-activity Tier 3 regions that dominate aggregate error, importance heavily favours autoregressive history, so recent local history is nearly sufficient where strikes are frequent and the multivariate models' advantage over ARIMA is concentrated in the sparser tiers.

The fine-tuned Chronos-2 ranks weather variables in its top seven, led by precipitation hours; this does not contradict the trees, since Chronos-2 receives no engineered lags and carries its autoregressive signal natively in the target channel, so permutation surfaces the strongest *exogenous* channel rather than re-discovering recent history. This meteorological prominence is consistent with the documented effect of weather on UAS and counter-UAS operations (Sobolewski, 2025).

Read together, the results situate the drone-strike case within known boundaries of conflict forecasting. Recent autoregressive history performing well even with a covariate rich structure, shows models anticipate escalations poorly (Bazzi et al., 2022; Hegre et al., 2021). With Chronos-2 performing competitively, it also highlights the promise of foundation models in conflict forecasting (Murphy et al., 2024).

Several aspects limit these findings. Methodologically, each configuration was evaluated only once. Repeated runs yielding confidence intervals would support firmer conclusions. Additionally, the feature-selection pipeline ranks on gain alone, whereas combining gain and permutation would prune features more reliably.

On the data side, distinguishing FPV from non-FPV strikes relies on unstructured text notes, introducing label noise. Furthermore, drone deployment intensified over the timeline, causing the training partition to under-represent the higher-intensity test set (Jensen & Atalan, 2025). While data volumes remain crucial, the rapid evolution of battlefield tactics (Molloy, 2024) means historical data quickly lose predictive relevance. Finally, strike activity is highly heterogeneous across regions, complicating global models.

Although every conflict has unique aspects, these limitations point to concrete next steps for conflict prediction more broadly. Graph neural networks could directly model cross-region dependence and heterogeneity (Li et al., 2023). Time-to-event models (Kiselev, 2023) would sidestep the zero inflation characteristic of the heavy-tailed, avalanche-like behavior of armed conflict (Cirillo & Taleb, 2016; Lee et al., 2020), while probabilistic forecasts could communicate the tail uncertainty that best supports operational decision-making.

9 CONCLUSION

This thesis evaluated machine- and deep-learning algorithms for the multi-horizon spatiotemporal forecasting of Russian drone strikes across Ukrainian regions. By comparing gradient-boosted trees, recurrent networks, and a time-series foundation model, this work contributes a reproducible dataset and benchmark for conflict-event forecasting.

Results show that multivariate models improve only marginally over a univariate ARIMA baseline (Sub-Q1). Short-term strike variance is driven predominantly by local autoregressive history and meteorological conditions, rather than broader geopolitical or macroeconomic covariates (Sub-Q3). While forecast quality for typical days remains stable across a seven-day horizon, the ability to anticipate high-intensity surges degrades significantly beyond four days (Sub-Q2). Furthermore, a single-stage Tweedie objective outperformed two-stage hurdle models, indicating that event occurrence and magnitude cannot be cleanly decoupled in this highly over-dispersed setting.

These findings align with broader conflict-forecasting research: baseline violence is predictable, but sharp, heavy-tailed deviations remain difficult to anticipate (Bazzi et al., 2022; Cirillo & Taleb, 2016). Operationally, near-term forecasts are dependable for routine tracking, but actionable lead times for escalations are limited to under four days. Model selection depends on priorities: foundation models like Chronos-2 excel at typical-day accuracy, while gradient-boosted trees better capture high-intensity days. Future research should target these rare surges using graph neural networks for cross-region dependencies (Li et al., 2023), time-to-event formulations (Kiselev, 2023), and probabilistic frameworks to communicate tail uncertainty for operational decision-making.

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A DRONE-STRIKE DISAMBIGUATION

ACLED records air-based offensive operations under a single event subtype, drawing no distinction between manned airstrikes and unmanned attacks, nor between short-range FPV systems and long-range loitering munitions such as the Shahed series. The notes field is therefore parsed in three stages to isolate drone strikes.

First, a preprocessing step neutralises phrases that mention drones without referring to drone strikes, principally reconnaissance-support clauses (“with the support of a recon drone”) and target descriptors such as “drone depot” or “UAV command”. These are replaced with placeholder tokens ([REMOVED_SUP_REC_DR], [REMOVED_TARGET_DRONE]), preserving traceability.

Second, regular-expression dictionaries are applied to the cleaned notes: an *airstrike* pattern matching manned-aviation terminology and munitions

(*Sukhoi*, *KAB/FAB/UPMB* designators, guided aerial bombs); an *offensive drone* pattern matching action verbs bound to drones together with named kamikaze platforms (*Shahed*, *Lancet*, *Lyuty*, *Molniya*); a general drone-mention pattern; and a *defensive context* pattern capturing interception language (*intercepted*, *shot down*, *debris*, *fragments*).

Lastly, the classification logic combines these signals to avoid actor / target / weapon mismatch. A record is labelled a drone strike if the offensive pattern fires, *or* if a general drone mention occurs in the absence of defensive context. Records describing only intercepted Shaheds or recovered UAV fragments are excluded from the positive class.

B CLASSIFICATION OF ACLED DAMAGE EVENTS

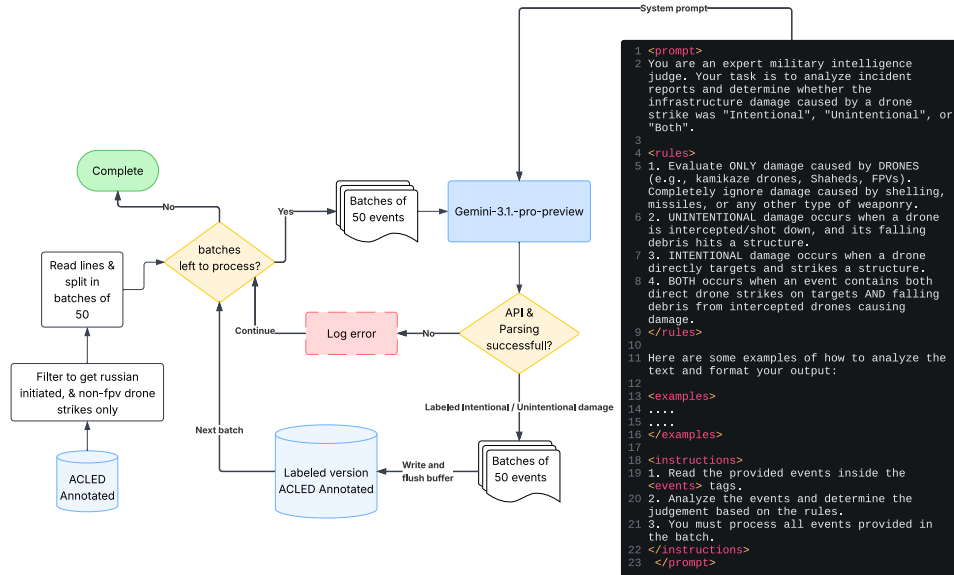


Figure 22: Batching pipeline implementing the LLM-as-a-judge framework. Filtered ACLED events are processed in batches of fifty, classified by Gemini-3.1-pro-preview as having caused intentional or unintentional damage, and written to the annotated output. Failed API or parsing calls are logged and the batch is retried.

The complete system prompt supplied to Gemini-3.1-pro-preview is reproduced verbatim in Listing 1. The instruction set, the four classification rules, and the five few-shot examples are shown exactly as sent; only batched event text was appended at inference time inside the `<events>` tags referenced at the end.

You are an expert military intelligence judge. Your task is to analyze incident reports and determine whether the infrastructure damage caused by a drone strike was "Intentional", "Unintentional", or "Both".

<rules>

1. Evaluate ONLY damage caused by DRONES (e.g., kamikaze drones, Shaheds, FPVs). Completely ignore damage caused by shelling, missiles, or any other type of weaponry.
2. UNINTENTIONAL damage occurs when a drone is intercepted/shot down, and its falling debris hits a structure.
3. INTENTIONAL damage occurs when a drone directly targets and strikes a structure.
4. BOTH occurs when an event contains both direct drone strikes on targets AND falling debris from intercepted drones causing damage.

</rules>

Here are some examples of how to analyze the text and format your output:

<examples>

<example>

<input>

Event_ID: 101

On 11 December 2023, Russian forces launched a Shahed-131/136 drone strike at Mykolaiv region (coded to Mykolaiv, Mykolaiv region). 13 drones were intercepted over Mykolaiv region, but the debris damaged residential houses in Ochakivska community (coded to Ochakiv, Mykolaiv region). On the same day, Russian forces shelled Ochakivska community, damaging residential houses. Casualties unknown.

</input>

<output>

101 | Unintentional | The report mentions damage from drone debris, which is unintentional. It also mentions damage from shelling, which must be ignored according to the rules. Therefore, the drone-related damage is purely unintentional.

</output>

</example>

<example>

<input>

Event_ID: 102

On 17 December 2023, Ukrainian forces shot down 7 Russian 'Shahed' drones over Kherson region (coded to Kherson). As a result of falling debris, 1 private house, 1 commercial office and a car service station were damaged. There were no casualties.

</input>

<output>

102 | Unintentional | The damage done was by falling debris and debris only. Meaning that the drone damage was unintentional.

</output>

</example>

<example>

<input>

Event_ID: 103

On 4 November 2024, Russian Forces launched two Kh-59/69 cruise missiles and 80 strike drones of Shahed and unidentified types at targets in Ukraine. Ukrainian Forces intercepted 50 drones, some of them over Odesa region. The fallen debris of the downed drones damaged 12 residential buildings in Bolhradaska community (coded to Bolhrad, Odesa). No casualties.

</input>

<output>

103 | Unintentional | Even though more drones were fired than intercepted, the only reported damage was done by the intercepted drones and their debris. Missiles are ignored. Thus, the damage was unintentional.

</output>

</example>

<example>

<input>

Event_ID: 104

On 6 June 2025, Russian drones and missiles struck an object of civilian infrastructure and an educational facility in Kyiv - Holosiivskiy, Kyiv City. Debris of intercepted Russian drones hit a gas station, damaging a car, and non-residential building. Casualties unknown.

</input>

<output>

104 | Both | The drones intentionally struck civilian infrastructure and an educational facility. Additionally, the debris of intercepted drones unintentionally hit a gas station and a non-residential building. This contains both types of drone damage.

</output>

</example>

<example>

<input>

Event_ID: 105

On 2 December 2025, Russian Forces carried out air strikes with 'KAB' aerial bombs, FPV-kamikaze-drones, dropped-drone borne explosive munitions and shelled (likely) with artillery/mortar, as well as MLRS at the bordering Velykopysarivska community (coded to Velyka Pysarivka, Sumy). Private houses were damaged. No casualties.

</input>

<output>

105 | Intentional | There are no mentions of debris hitting buildings. There is an intentional strike utilizing FPV-Kamikaze-drones and dropped munitions resulting in damaged houses. Other weapons are ignored. This is intentional drone damage.

</output>

</example>

</examples>

<instructions>

1. Read the provided events inside the <events> tags.
2. Analyze the events and determine the judgement based on the rules.
3. You must process all events provided in the batch.

</instructions>

Listing 1: System prompt used for LLM-as-a-judge classification of ACLED drone-damage events (Gemini-3.1-pro-preview), reproduced verbatim.

C HYPERPARAMETER TUNING DETAILS

This appendix reports the Optuna search spaces and the best configurations recovered for each of the top-five models in Table 5. All studies used the TPE sampler with seed 42 and a MedianPruner. In the GBDT studies no trial was pruned; in the LSTM-Poisson- $w=28$ study, 25 of 50 trials were pruned before completion.

Training, hyperparameter tuning, and evaluation ran on an NVIDIA A40 (Tilburg University GPU4EDU cluster) for the deep-learning models, a T4 (Google Colab) for the GBDTs, and an Intel Core i7-11800H for the hurdle model and statistical baselines.

Table 7: Optuna search spaces per model family.

Family	Hyperparameter	Range / set
LightGBM	num_leaves	[15, 127] (int)
	max_depth	[3, 10] (int)
	min_child_samples	[10, 200] (int)
	learning_rate	$[10^{-2}, 0.2]$ (log)
	n_estimators	{200, 300, ..., 1000}
	subsample	[0.6, 1.0]
	colsample_bytree	[0.6, 1.0]
	reg_alpha, reg_lambda	$[10^{-3}, 10]$ (log)
	tweedie_variance_power	[1.1, 1.9] (Tweedie only)
XGBoost	max_depth	[3, 10] (int)
	min_child_weight	[1, 10] (int)
	learning_rate	$[10^{-2}, 0.2]$ (log)
	n_estimators	{200, 300, ..., 1000}
	subsample	[0.6, 1.0]
	colsample_bytree	[0.6, 1.0]
	reg_alpha, reg_lambda	$[10^{-3}, 10]$ (log)
	tweedie_variance_power	[1.1, 1.9] (Tweedie only)
CatBoost	depth	[4, 8] (int)
	learning_rate	$[10^{-2}, 0.2]$ (log)
	iterations	{200, 300, ..., 1000}
	l2_leaf_reg	{1.0, 1.5, ..., 5.0}
	subsample	[0.6, 1.0]
	tweedie_variance_power	[1.1, 1.9] (Tweedie only)
LSTM / GRU	hidden_dim	{16, 32, 64, 128}
	n_rnn_layers	[1, 3] (int)
	hidden_fc_sizes	{none, [32], [64], [64, 32]}
	dropout	[0.0, 0.4]
	batch_size	{32, 64, 128, 256}
	lr	$[10^{-4}, 10^{-2}]$ (log)
	weight_decay	$[10^{-6}, 10^{-2}]$ (log)
	tweedie_variance_power	[1.1, 1.9] (Tweedie only)
Chronos-2	fine_tune_lr	$[10^{-6}, 10^{-4}]$ (log)
	fine_tune_steps	{600, 700, ..., 3000}

Table 8: Best configurations for the top-five models in Table 5. CatBoost-Tweedie and LSTM-Poisson configurations were obtained under the Global paradigm and reused under the Activity paradigm.

Model	Hyperparameter	Value
CatBoost-Tweedie	depth	4
	learning_rate	0.01207
	iterations	900
	l2_leaf_reg	5.0
	subsample	0.998
	tweedie_variance_power	1.395
LightGBM-Poisson	num_leaves	15
	max_depth	8
	min_child_samples	168
	learning_rate	0.01221
	n_estimators	700
	subsample	0.797
	colsample_bytree	0.826
	reg_alpha	0.104
LSTM-Poisson ($w=28$)	reg_lambda	0.642
	hidden_dim	64
	n_rnn_layers	1
	hidden_fc_sizes	[64]
	dropout	0.251
	batch_size	32
	lr	1.081×10^{-3}
Chronos-2 (fine-tuned)	weight_decay	4.960×10^{-4}
	fine_tune_lr	8.72×10^{-5}
	fine_tune_steps	1500

Table 9: All Base Gathered Variables

Variable	Variable
env_rus_holiday	env_ua_holiday
env_weather_wind_direction_10m_dominant	env_weather_eto_fao_evapotranspiration
env_weather_rain_sum	env_weather_cloud_cover_min
env_weather_wind_speed_10m_mean	env_weather_cloud_cover_max
env_weather_daylight_duration	env_k_max
env_weather_temperature_2m_mean	env_weather_sunshine_duration
env_weather_pressure_msl_min	env_weather_soil_moisture_o_to_100cm_mean
env_weather_snowfall_sum	env_weather_precipitation_hours

Continued on next page

Table 9 – continued from previous page

Variable	Variable
env_weather_pressure_msl_mean	env_weather_shortwave_radiation_sum
env_weather_temperature_2m_max	env_weather_wind_gusts_10m_mean
env_weather_precipitation_sum	env_weather_temperature_2m_min
env_weather_pressure_msl_max	env_weather_cloud_cover_mean
env_weather_soil_temperature_o_to_100cm_mean	env_weather_wind_gusts_10m_max
env_weather_wind_speed_10m_max	acled_other_rus_armed_clash
acled_other_rus_disrupted_weapons_use	acled_other_rus_shelling_artillery_missile_attack
acled_other_ua_armed_clash	acled_other_ua_disrupted_weapons_use
acled_other_ua_shelling_artillery_missile_attack	act_drone_infra_ua_education_intent
act_drone_infra_ua_education_unintent	act_drone_infra_ua_energy_intent
act_drone_infra_ua_energy_unintent	act_drone_infra_ua_health_intent
act_drone_infra_ua_health_unintent	act_drone_infra_ua_residential_intent
act_drone_infra_ua_residential_unintent	act_drone_strike_on_rus_in_ua_nofpv
dist_to_nearest_contested_km	dist_to_nearest_ru_km
act_confirmed_launched	act_confirmed_destroyed
act_intercept_rate	act_cyber_rus_ua
act_cyber_rus_nato	act_cyber_rus_other
act_cyber_ua_rus	act_cyber_ua_other
act_cyber_nato_rus	act_cyber_nato_other
act_cyber_other_nato	act_cyber_other_rus
act_cyber_other_ua	fin_ch_export_belarus
fin_ch_export_dprk	fin_ch_export_iran
fin_ch_export_kazakhstan	fin_ch_export_kyrgyzstan
fin_ch_export_russia	fin_ch_export_tajikistan
fin_ch_export_turkmenistan	fin_ch_export_uzbekistan
total_contested_area_km2	total_ru_area_km2
com_aid_financial	com_aid_humanitarian
com_aid_military	fin_usd_rub_rate
fin_usd_uah_rate	com_diplo_rus_ua
com_escalationsignal_rus_ua	com_restrictive_rus_ua
com_verbal_conflict_rus_ua	com_ners_r1
com_ners_r2	com_ners_r3
com_ners_r4	com_ners_r5
com_ners_r6	com_ners_r7
com_ners_r8	com_ners_w1
com_ners_w2	com_ners_w3
com_ners_w4	com_ners_w5
com_ners_w6	com_ners_w7

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Table 9 – continued from previous page

Variable	Variable
com_ners_w8	act_drone_strike_on_rus_in_rus_ nofpv
act_ua_specialmilitaryoperation	act_total_daily_strike_events
act_total_damage_events	ratio_ua_rus_armed_clash
dist_x_clash	com_restrictive_redsecondary_ua
com_verbal_conflict_bluesecondary_ rus	com_diplo_bluesecondary_rus
com_verbal_conflict_redsecondary_ua	com_logistic_redsecondary_rus
com_restrictive_redsecondary_ bluesecondary	com_diplo_redsecondary_ua
com_verbal_conflict_redsecondary_ rus	com_escalationsignal_bluesecondary_ rus
com_diplo_redsecondary_rus	com_escalationsignal_redsecondary_ ua
com_diplo_bluesecondary_ua	com_restrictive_bluesecondary_rus

E TOP 20 MODELS

Table 10: Top 20 Models, Evaluation Metrics

Modelname	paradigm	model	SkillScore	mae	rmse
gbdt	activity	catboost_tweedie_tuned	0.169634	0.767909	1.830539
chronos2	local	chronos2_fine_tuned	0.169486	0.713611	1.830864
gbdt	global	catboost_tweedie_tuned	0.169411	0.785818	1.831029
gbdt	global	lightgbm_poisson_tuned	0.165593	0.775707	1.839445
lstm	activity	lstm_poisson_w28_tuned	0.165131	0.777548	1.840464
chronos2	local	chronos2_zero_shot	0.161110	0.719448	1.849330
diff	global	arima	0.159221	0.813287	1.853493
gbdt	activity	lightgbm_poisson_tuned	0.159210	0.766848	1.853518
gbdt	activity	catboost_poisson_tuned	0.153791	0.779569	1.865464
gbdt	global	xgboost_tweedie_tuned	0.152136	0.778661	1.869111
lstm	activity	gru_poisson_w28_tuned	0.151309	0.782510	1.870936
gbdt	activity	xgboost_tweedie_tuned	0.150523	0.764992	1.872668
lstm	activity	gru_poisson_w14_tuned	0.149851	0.781645	1.874149
gbdt	global	xgboost_poisson_tuned	0.149798	0.790820	1.874267
gbdt	activity	xgboost_poisson_tuned	0.148924	0.774119	1.876192
gbdt	local	lightgbm_poisson_tuned	0.147808	0.783284	1.878654
gbdt	global	catboost_poisson_tuned	0.145142	0.804140	1.884530
lstm	activity	lstm_tweedie_w28_tuned	0.144459	0.792681	1.886036
lstm	global	gru_poisson_w14_tuned	0.144243	0.799084	1.886513
lstm	global	lstm_poisson_w28_tuned	0.143140	0.799503	1.888945

F FEATURE IMPORTANCE

Table 11: Semantic category mapping rules for feature importance separation.

Semantic Category	Matching Feature Substrings
Autoregressive strikes	total_daily_strike_events, drone_strike_on, _target, specialmilitary, ratio_ua_rus
Spatial / static	region_statcov, dist_to_nearest, dist_x_clash, area_km2
Weather / geomag.	env_weather, env_k_max
Conflict & damage	disrupted_weapons, shelling, armed_clash, damage_events, drone_infra, acled_other
Macroeconomic	usd_rub, usd_uah, ch_export
Comms / diplo / aid	com_aid, com_diplo, com_verbal, com_restrictive, com_escalation, com_logistic, com_ners
Missile / launch	confirmed_launched, confirmed_destroyed, intercept_rate
Cyber	act_cyber
Calendar	holiday
Other	<i>Any feature not matching the above patterns</i>

Top 15 Feature Importances per Activity Level



Figure 23: CatBoost feature importance (Information Gain and Permutation) per activity tier.